

E255890625_2

The equation $x^4 + x + y^3 + 6y - 1 = 0$ has no integer solutions

Size $h = 39$, length $l = 11.58$

Theorem 1. *The equation*

$$x^4 + x + y^3 + 6y - 1 = 0 \tag{1}$$

has no integer solutions.

The proof uses 47 auxiliary polynomials $H_0, \dots, H_{46}$, 7 of which are constants, and a graph on these vertices with 125 edges; to each edge $\{i, j\}$ we attach the Hilbert symbol $(H_i, H_j)_v$ of the values at a hypothetical solution. By the product formula, the product of these symbols over all edges and all places is 1. We show that the product over the edges is 1 at every odd prime outside a finite set S , compute it at the primes of S and at the real place, and obtain a contradiction. Apart from polynomial identities, which can be checked by expanding both sides, and finite checks of congruences, the proof uses only basic properties of the Legendre symbol and of the Hilbert symbol.

1 Legendre symbols and Hilbert symbols

For a prime p and a non-zero integer N , we write $v_p(N)$ for the exponent of p in the prime factorization of N . For an integer a and an odd prime p , the Legendre symbol (a/p) is 0 if $p \mid a$, 1 if a is a non-zero square modulo p , and -1 otherwise. It is multiplicative in a , and it depends only on a modulo p [1, Chapter 5].

For a prime p , let $\mathbb{Q}_p$ be the field of p -adic numbers, and put $\mathbb{Q}_\infty = \mathbb{R}$. For v a prime or $v = \infty$, and non-zero $a, b \in \mathbb{Q}$, the Hilbert symbol $(a, b)_v$ is 1 if the equation $z^2 = ar^2 + bs^2$ has a solution $(r, s, z) \neq (0, 0, 0)$ in $\mathbb{Q}_v$, and -1 otherwise. In particular, $(a, b)_v = 1$ for every v if this equation has a non-zero solution in $\mathbb{Q}$. For an odd integer t , we put $\varepsilon(t) = (t - 1)/2$ and $\omega(t) = (t^2 - 1)/8$. We use the following facts [2, Chapter III, Proposition 2 and Theorems 1 and 3].

Theorem 2. *Let a, a' and b be non-zero integers, and let v be a prime or $v = \infty$.*

- (i) $(a, b)_\infty = -1$ if and only if $a < 0$ and $b < 0$.
- (ii) Let p be an odd prime, and write $a = p^\alpha u$ and $b = p^\beta w$ with $p \nmid uw$. Then $(a, b)_p = (-1)^{\alpha\beta\varepsilon(p)}(u/p)^\beta(w/p)^\alpha$.
- (iii) Write $a = 2^\alpha u$ and $b = 2^\beta w$ with u and w odd. Then $(a, b)_2 = (-1)^{\varepsilon(u)\varepsilon(w) + \alpha\omega(w) + \beta\omega(u)}$.
- (iv) $(a, b)_v = 1$ for all but finitely many v , and $\prod_v (a, b)_v = 1$, the product over all primes v and $v = \infty$.
- (v) $(a, b)_v = (b, a)_v$ and $(aa', b)_v = (a, b)_v(a', b)_v$.

Lemma 3. *Let a and b be non-zero integers. If p is an odd prime with $p \nmid a$ and $(a/p) = 1$, then $(a, b)_p = 1$. If $a \equiv 1 \pmod{8}$, then $(a, b)_2 = 1$.*

Proof. The first claim is Theorem 2(ii) with $\alpha = 0$. For the second, $\varepsilon(a)$ and $\omega(a)$ are even, and Theorem 2(iii) with $\alpha = 0$ gives $(a, b)_2 = 1$. $\square$

Lemma 4. *Let p be a prime. For a non-zero integer X , write $X = p^j u$ with $p \nmid u$. If p is odd, put $c_p(X) = (j, e) \in \mathbb{F}_2^2$, where $(u/p) = (-1)^e$; if $p = 2$, put $c_2(X) = (j, e_1, e_2) \in \mathbb{F}_2^3$, where $u \equiv (-1)^{e_1} 5^{e_2} \pmod{8}$; all entries are taken modulo 2. Then $c_p(XX') = c_p(X) + c_p(X')$ and $(X, X')_p = (-1)^{\beta_p(c_p(X), c_p(X'))}$ for non-zero integers X, X' , where β_p is the symmetric bilinear form*

$$\begin{aligned} \beta_p((j, e), (j', e')) &= jj' \varepsilon(p) + je' + j'e && \text{if } p \text{ is odd,} \\ \beta_2((j, e_1, e_2), (j', e'_1, e'_2)) &= e_1 e'_1 + je'_2 + j'e_2. \end{aligned}$$

Proof. For odd p , this is Theorem 2(ii), as the Legendre symbol is multiplicative. For $p = 2$, the odd residues 1, 3, 5, 7 modulo 8 are $(-1)^{e_1} 5^{e_2}$ with $(e_1, e_2) = (0, 0), (1, 1), (0, 1), (1, 0)$, respectively; in each case $\varepsilon(u) \equiv e_1$ and $\omega(u) \equiv e_2 \pmod{2}$, so Theorem 2(iii) gives the formula, and c_2 is additive because $(e_1, e_2) \mapsto (-1)^{e_1} 5^{e_2}$ is an isomorphism from $\mathbb{F}_2^2$ to the group of odd residues modulo 8. $\square$

Lemma 5. *Let p be a prime, $k \geq 1$, and let X be a non-zero integer with $X \equiv r \pmod{p^k}$, where $0 \leq r < p^k$. Then $c_p(X) \in b + D$ for the vector b and the subspace D given as follows. If $r = 0$, then $b = 0$ and D is the whole space. Otherwise write $r = p^j r'$ with $p \nmid r'$, so that $j < k$. If p is odd, or if $p = 2$ and $k - j \geq 3$, then $b = c_p(r)$ and $D = 0$. If $p = 2$ and $k - j = 2$, then $b = (j, e_1, 0)$ with $r' \equiv (-1)^{e_1} \pmod{4}$, and D is spanned by $(0, 0, 1)$. If $p = 2$ and $k - j = 1$, then $b = (j, 0, 0)$, and D is spanned by $(0, 1, 0)$ and $(0, 0, 1)$.*

Proof. If $r \neq 0$, then $v_p(X) = j$ and $X/p^j \equiv r' \pmod{p^{k-j}}$. For odd p , the Legendre symbol of X/p^j is therefore that of r' . For $p = 2$, the residue of $X/2^j$ modulo 8 is that of r' if $k - j \geq 3$, and its residue modulo 4, which determines e_1 , is that of r' if $k - j = 2$. $\square$

Lemma 6. *Let p be a prime, and let E be a set of edges on a finite set of vertices. For every vertex i , let b_i be a vector and D_i a subspace of $\mathbb{F}_2^2$ (of $\mathbb{F}_2^3$ if $p = 2$), and put $\sigma_i = \sum_{j \sim i} b_j$, the sum over the edges $\{i, j\} \in E$. Suppose that $\beta_p(d, \sigma_i) = 0$ for every vertex i and every $d \in D_i$, and that $\beta_p(d, d') = 0$ for every edge $\{i, j\} \in E$ and all $d \in D_i$ and $d' \in D_j$. Then*

$$\sum_{\{i, j\} \in E} \beta_p(c_i, c_j) = \sum_{\{i, j\} \in E} \beta_p(b_i, b_j) \quad \text{whenever } c_i \in b_i + D_i \text{ for every } i.$$

By bilinearity, it suffices to check the hypotheses for d and d' in spanning sets of D_i and D_j .

Proof. Write $c_i = b_i + d_i$ with $d_i \in D_i$. As β_p is bilinear and symmetric, the left side equals the right side plus $\sum_i \beta_p(d_i, \sigma_i) + \sum_{\{i, j\} \in E} \beta_p(d_i, d_j)$, and both sums vanish. $\square$

Lemma 7. *Let $f(z) = az^3 + bz^2 + cz + d$ be a real polynomial with $a \neq 0$ and discriminant $\Delta < 0$. Then f has exactly one real root η , and $f(z)$ has the sign of $a(z - \eta)$ for every real $z \neq \eta$.*

Proof. The discriminant is $\Delta = a^4 \prod_{i < j} (\rho_i - \rho_j)^2$, where ρ_1, ρ_2, ρ_3 are the complex roots of f . If all roots were real, then $\Delta \geq 0$. Hence f has a non-real root ρ ; then $\bar{\rho} \neq \rho$ is also a root, and the third root η is real. Thus $f(z) = a(z - \eta)(z - \rho)(z - \bar{\rho}) = a(z - \eta)|z - \rho|^2$ for real z , and $|z - \rho|^2 > 0$. $\square$

2 The graph

Put $F(x, y) = x^4 + x + y^3 + 6y - 1$, so that (1) is the equation $F(x, y) = 0$. The vertices are the polynomials

$$\begin{aligned} H_0 &= -y, \\ H_1 &= -x - y, \\ H_2 &= -y - 1, \\ H_3 &= -y + 1, \\ H_4 &= x - y, \\ H_5 &= -2x - y, \\ H_6 &= -x - y - 1, \\ H_7 &= -x - y + 1, \\ H_8 &= -y - 2, \\ H_9 &= x - y - 1, \\ H_{10} &= 2x - y, \\ H_{11} &= -2x - y - 1, \\ H_{12} &= -2x - y + 1, \\ H_{13} &= -x - y - 2, \\ H_{14} &= x - y - 2, \\ H_{15} &= x - y + 2, \\ H_{16} &= 2x - y + 1, \\ H_{17} &= -2x - y + 2, \\ H_{18} &= 2x - y - 2, \\ H_{19} &= x^2 + y, \\ H_{20} &= x^2 - y, \\ H_{21} &= x^2 + x + y, \\ H_{22} &= x^2 - x - y, \\ H_{23} &= x^2 - y - 1, \\ H_{24} &= x^2 - y + 1, \\ H_{25} &= x^2 + x - y, \\ H_{26} &= x^2 - x + y - 1, \\ H_{27} &= x^2 - x + y + 1, \\ H_{28} &= x^2 + y + 2, \\ H_{29} &= x^2 + x + y - 1, \\ H_{30} &= x^2 + 2x + y, \\ H_{31} &= x^2 - x - y - 1, \\ H_{32} &= x^2 - x - y + 1, \\ H_{33} &= x^2 - y - 2, \\ H_{34} &= x^2 + x - y - 1, \\ H_{35} &= x^2 - 2x + y - 1, \\ H_{36} &= x^2 - 2x + y + 1, \end{aligned}$$

$$\begin{aligned}
H_{37} &= x^2 - x + y + 2, \\
H_{38} &= x^2 + 2x + y - 1, \\
H_{39} &= x^2 + 2x + y + 1, \\
H_{40} &= 1, \\
H_{41} &= -1, \\
H_{42} &= 2, \\
H_{43} &= 3, \\
H_{44} &= 15, \\
H_{45} &= 6, \\
H_{46} &= -2.
\end{aligned}$$

The edges are $\{0, 12\}, \{0, 26\}, \{0, 28\}, \{0, 37\}, \{1, 4\}, \{1, 8\}, \{1, 11\}, \{1, 24\}, \{1, 25\}, \{1, 32\}, \{1, 33\}, \{2, 3\}, \{2, 4\}, \{2, 14\}, \{2, 15\}, \{2, 19\}, \{2, 22\}, \{2, 32\}, \{2, 33\}, \{2, 36\}, \{2, 39\}, \{3, 10\}, \{3, 19\}, \{3, 20\}, \{3, 22\}, \{3, 27\}, \{3, 29\}, \{3, 34\}, \{3, 35\}, \{3, 36\}, \{4, 19\}, \{4, 28\}, \{4, 29\}, \{4, 34\}, \{4, 39\}, \{5, 28\}, \{5, 39\}, \{6, 19\}, \{6, 22\}, \{6, 23\}, \{6, 26\}, \{6, 36\}, \{7, 19\}, \{7, 31\}, \{7, 39\}, \{8, 12\}, \{8, 15\}, \{8, 24\}, \{8, 31\}, \{9, 12\}, \{9, 24\}, \{9, 36\}, \{9, 39\}, \{10, 21\}, \{10, 23\}, \{10, 28\}, \{10, 29\}, \{10, 36\}, \{11, 15\}, \{12, 15\}, \{12, 31\}, \{12, 37\}, \{12, 39\}, \{13, 17\}, \{13, 21\}, \{13, 24\}, \{13, 26\}, \{13, 35\}, \{14, 24\}, \{14, 32\}, \{14, 35\}, \{15, 22\}, \{15, 26\}, \{15, 32\}, \{15, 39\}, \{16, 21\}, \{16, 26\}, \{16, 27\}, \{16, 28\}, \{16, 37\}, \{17, 23\}, \{17, 27\}, \{17, 29\}, \{17, 35\}, \{17, 37\}, \{18, 23\}, \{19, 35\}, \{20, 31\}, \{20, 34\}, \{21, 35\}, \{22, 25\}, \{22, 29\}, \{22, 32\}, \{23, 29\}, \{25, 33\}, \{26, 27\}, \{26, 28\}, \{27, 32\}, \{28, 32\}, \{29, 38\}, \{30, 39\}, \{31, 35\}, \{31, 39\}, \{33, 34\}, \{34, 36\}, \{35, 38\}, \{1, 41\}, \{2, 41\}, \{3, 42\}, \{4, 43\}, \{5, 44\}, \{6, 42\}, \{7, 42\}, \{11, 43\}, \{13, 42\}, \{15, 45\}, \{16, 42\}, \{19, 41\}, \{21, 46\}, \{25, 46\}, \{29, 42\}, \{31, 41\}, \{34, 46\}, \{35, 41\}, \{36, 41\}. For a vertex i , we write $j \sim i$ if $\{i, j\}$ is an edge. For every non-constant vertex i , Table 1 gives a positive integer m_i and polynomials W_i , A_i and B_i with integer coefficients such that$

$$m_i^2 \prod_{j \sim i} H_j - W_i^2 = A_i F + B_i H_i. \quad (2)$$

For every edge $\{i, j\}$ between two non-constant vertices, Table 2 gives a positive integer k_{ij} and polynomials α_{ij} , β_{ij} and γ_{ij} with integer coefficients such that

$$\alpha_{ij} F + \beta_{ij} H_i + \gamma_{ij} H_j = k_{ij}. \quad (3)$$

These identities can be checked by expanding both sides. For the signs of the vertices, we regard F as a polynomial in y ; it has degree 3 and leading coefficient 1, and its discriminant is

$$\Delta(x) = -27x^8 - 54x^5 + 54x^4 - 27x^2 + 54x - 891.$$

For $i \in I = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39\}$, the vertex H_i has the form $k_i y + r_i(x)$ with a non-zero integer k_i , and Table 3 lists the polynomial $\varphi_i(x) = k_i^3 F(x, -r_i(x)/k_i)$, which has integer coefficients.

3 Proof of Theorem 1

Suppose that $(x, y) \in \mathbb{Z}^2$ is a solution of (1), so that $F(x, y) = 0$; from now on the polynomials denote their values at (x, y) .

Step 1: the values are non-zero. The non-constant H_i are non-zero by Step 2.

Step 2: signs. *The signs of H_i for $i \in I$.* If $\Delta(x) < 0$, then by Lemma 7, applied to $F(x, z)$ as a polynomial in z , y is its only real root, and $F(x, z)$ has the sign of $(z - y)$ for $z \neq y$. For $i \in I$ and $z = -r_i(x)/k_i$, we have $z - y = -H_i/k_i$, so if moreover $\varphi_i(x) = k_i^3 F(x, z) \neq 0$, then $H_i \neq 0$, and H_i has the sign of $-\varphi_i(x)$. Expanding $\Delta(10 + t)$ and the $\varphi_i(10 + t)$ as polynomials in t gives in each case coefficients of one sign and a non-zero constant term, and likewise for $\Delta(-11 - t)$ and the $\varphi_i(-11 - t)$. Hence, for $x \geq 10$ and for $x \leq -11$, we have $\Delta(x) < 0$, and $\varphi_i(x)$ is non-zero and has the sign of the constant term of the corresponding expansion; the resulting signs of the H_i are listed in Table 3. It remains to consider $-10 \leq x \leq 9$. Checking the residues of y , we find that for $x \in \{-10, -9, -6, -5, -4, -1, 0, 1, 4, 5, 6, 9\}$, we have $F(x, y) \not\equiv 0 \pmod{5}$ for every y , for $x \in \{-8, -7, 2, 3\}$, we have $F(x, y) \not\equiv 0 \pmod{11}$ for every y , for $x = 7$, we have $F(x, y) \not\equiv 0 \pmod{13}$ for every y , for $x = -3$ and $x = -2$, we have $F(x, y) \not\equiv 0 \pmod{17}$ for every y , and for $x = 8$, we have $F(x, y) \not\equiv 0 \pmod{23}$ for every y . Hence (1) has no solution with $-10 \leq x \leq 9$.

For a place v , put $B_v = \prod_{\{i,j\}} (H_i, H_j)_v$, the product over the edges. By Theorem 2(i), $(H_i, H_j)_\infty = -1$ exactly when $H_i < 0$ and $H_j < 0$. Here $H_i > 0$ for $i \in \{40, 42, 43, 44, 45\}$ and $H_i < 0$ for $i \in \{41, 46\}$. The other signs are given in Table 3 for $x \geq 10$ and for $x \leq -11$. In both cases, counting the edges whose two values are negative gives $B_\infty = 1$.

Step 3: odd primes outside S . Let $S = \{2, 3, 5, 7, 11, 13, 17\}$ be the set of primes dividing 2, the m_i , the k_{ij} and the constant vertices. Let $p \notin S$ be an odd prime; we show that $B_p = 1$. For an edge $\{i, j\}$ such that p divides neither H_i nor H_j , $(H_i, H_j)_p = 1$ by Theorem 2(ii). Now let i be a vertex with $p \mid H_i$. Then H_i is not constant, as $p \notin S$. For every edge $\{i, j\}$, $p \nmid H_j$: if H_j is constant, because $p \notin S$, and otherwise by (3), because $p \nmid k_{ij}$. In particular, no edge has both values divisible by p . By Theorem 2(v), the edges at i contribute $\prod_{j \sim i} (H_i, H_j)_p = (H_i, \Pi_i)_p$, where $\Pi_i = \prod_{j \sim i} H_j$ is not divisible by p . By (2), $m_i^2 \Pi_i \equiv W_i^2 \pmod{p}$, and $p \nmid m_i$, so Π_i is a non-zero square modulo p , and $(H_i, \Pi_i)_p = (\Pi_i, H_i)_p = 1$ by Lemma 3. Hence $B_p = 1$.

Step 4: the primes in S . Let $p \in S$. By Lemma 4, $B_p = (-1)^N$, where $N = \sum_{\{i,j\}} \beta_p(c_p(H_i), c_p(H_j))$, the sum over the edges. Suppose that $x \equiv a$ and $y \equiv b \pmod{p^k}$. Then $H_i \equiv H_i(a, b) \pmod{p^k}$, and Lemma 5 gives a vector b_i and a subspace D_i with $c_p(H_i) \in b_i + D_i$; for a constant vertex, we take $b_i = c_p(H_i)$ and $D_i = 0$. If the hypotheses of Lemma 6 hold for the edges, then $N = \sum_{\{i,j\}} \beta_p(b_i, b_j)$, so B_p is determined by a , b and k . Starting from the solutions of (1) modulo p , we refine a residue class $x \equiv a$, $y \equiv b \pmod{p^k}$ to the classes modulo p^{k+1} that contain solutions of (1) modulo p^{k+1} , until these hypotheses hold, or until no solution remains. Table 4 lists the resulting terminal classes together with the value of B_p ; every integer solution lies in one of them. The result is $B_2 = 1$, $B_3 = 1$, $B_5 = -1$, $B_7 = 1$, $B_{11} = 1$, $B_{13} = 1$, $B_{17} = 1$.

Step 5: contradiction. By Theorem 2(iv), applied to each edge, $\prod_v B_v = 1$, the product over all primes and $v = \infty$. By Step 3, the factors with $p \notin S$ are 1, so $\prod_{p \in S} B_p = B_\infty = 1$ by Step 2. But by Step 4, $\prod_{p \in S} B_p = -1$. This contradiction proves the theorem. $\square$

References

- [1] K. Ireland and M. Rosen, *A Classical Introduction to Modern Number Theory*, 2nd ed., Graduate Texts in Mathematics 84, Springer, New York, 1990.
- [2] J.-P. Serre, *A Course in Arithmetic*, Graduate Texts in Mathematics 7, Springer, New York, 1973.

Table 1: The data of the identities (2).

$i = 0, m = 1$, neighbours 12, 26, 28, 37	
W	$2x^3 - 2x^2 + x - 2$
A	$-2x^3 - x^2y + x^2 - 4xy - 2x - 3y^2 + 7y + 8$
B	$-2x^3y^2 + 2x^3y - x^2y^3 + 4x^2y^2 - 6x^2y - 12x^2 - 4xy^3 - 2xy^2 - 24xy$ $+ 2x - 3y^4 + 8y^3 - 8y^2 + 42y + 37$
$i = 1, m = 1$, neighbours 41, 4, 8, 11, 24, 25, 32, 33	
W	$6x^3 - 2x^2 + 6x + 2$
A	$-2x^6y - 4x^6 + x^5y^2 + x^5y - 2x^5 + x^4y^3 + 11x^4y^2 + 20x^4y + 4x^4$ $- 4x^3y^3 - 5x^3y^2 + 7x^3y + 2x^3 - 2x^2y^4 - 21x^2y^3 - 27x^2y^2$ $+ 16x^2y - 32x^2 - xy^5 + 5xy^4 + 4xy^3 - 21xy^2 - xy + 26x - y^6 - 7y^5 + 30y^4$ $- 47y^3 + 23y^2 + 22y + 4$
B	$-2x^3y^5 + 28x^3y^4 - 14x^3y^3 + 138x^3y^2 + 54x^3y + 76x^3$ $- 2x^2y^6 - 29x^2y^5 + 2x^2y^4 - 186x^2y^3 - 28x^2y^2 - 43x^2y$ $- 42x^2 + 12xy^6 - 27xy^5 + 92xy^4 - 162xy^3 + 130xy^2 - 155xy + 82x - y^8 - 7y^7$ $+ 23y^6 - 91y^5 + 211y^4 - 283y^3 + 189y^2 + 105y + 2$
$i = 2, m = 1$, neighbours 41, 3, 4, 14, 15, 19, 22, 32, 33, 36, 39	
W	$-16x^3 - 12x^2 + 36x + 64$
A	$x^{11}y - x^{11} - 3x^{10}y^2 + x^{10}y + 2x^{10} + 3x^9y^3 + 3x^9y^2 - 12x^9y$ $+ 6x^9 - x^8y^4 - 5x^8y^3 + 14x^8y^2 + 6x^8y - 14x^8 + x^7y^4$ $- 4x^7y^3 - 22x^7y^2 + 33x^7y - 8x^7 + 3x^6y^5 + 4x^6y^4 + 12x^6y^3$ $- 9x^6y^2 - 40x^6y + 30x^6 - 3x^5y^6 - 10x^5y^5 + 4x^5y^4 - 19x^5y^3$ $+ 66x^5y^2 - 38x^5y + x^4y^7 + 8x^4y^6 - 6x^4y^5 + 12x^4y^4$ $- 30x^4y^3 - 50x^4y^2 + 94x^4y - 29x^4 - x^3y^7 + 4x^3y^5 - 23x^3y^4$ $+ 126x^3y^3 - 157x^3y^2 + 53x^3y - 2x^3 - 3x^2y^8 - 4x^2y^7$ $- 22x^2y^6 + 17x^2y^5 - 32x^2y^4 - 5x^2y^3 + 189x^2y^2 - 164x^2y$ $- 232x^2 + 3xy^9 + 10xy^8 + 18xy^7 + 57xy^6 - 112xy^5 + 140xy^4 - 305xy^3$ $+ 292xy^2 - 116xy - 371x - y^{10} - 8y^9 - 3y^8 - 57y^7 + 84y^6 - 28y^5 + 107y^4$ $+ 120y^3 - 416y^2 + 221y + 989$
B	$-x^3y^9 + x^3y^8 - 8x^3y^7 - 31x^3y^6 + 160x^3y^5 - 413x^3y^4$ $+ 1174x^3y^3 - 2253x^3y^2 + 2929x^3y - 3142x^3 - 3x^2y^{10} - x^2y^9$ $- 36x^2y^8 + 49x^2y^7 - 180x^2y^6 + 321x^2y^5 - 484x^2y^4$ $+ 518x^2y^3 + 88x^2y^2 - 973x^2y - 379x^2 + 3xy^{11} + 7xy^{10} + 28xy^9$ $+ 79xy^8 - 95xy^7 + 501xy^6 - 1444xy^5 + 2668xy^4 - 4655xy^3 + 6453xy^2 - 7857xy$ $+ 5968x - y^{12} - 7y^{11} - 2y^{10} - 103y^9 + 175y^8 - 536y^7 + 1216y^6 - 1357y^5$ $+ 1593y^4 - 755y^3 - 864y^2 + 2606y + 3107$
$i = 3, m = 1$, neighbours 42, 2, 10, 19, 20, 22, 27, 29, 34, 35, 36	
W	$44x^3 - 8x^2 - 124x + 216$
A	$-4x^{13}y - 4x^{13} + 2x^{12}y^2 + 18x^{12}y + 16x^{12} - 16x^{11}y^2 - 20x^{11}y$ $- 4x^{11} + 4x^{10}y^3 + 22x^{10}y^2 - 38x^{10}y - 56x^{10} + 4x^9y^4 + 4x^9y^3$ $+ 80x^9y^2 + 156x^9y + 76x^9 - 2x^8y^5 - 22x^8y^4 - 94x^8y^3$ $- 280x^8y^2 - 198x^8y + 8x^8 + 16x^7y^5 + 68x^7y^4 + 244x^7y^3$ $+ 208x^7y^2 - 56x^7y - 72x^7 - 4x^6y^6 - 34x^6y^5 - 80x^6y^4$ $- 144x^6y^3 + 338x^6y^2 + 460x^6y + 28x^6 - 4x^5y^7 - 4x^5y^6$ $- 78x^5y^5 - 160x^5y^4 - 616x^5y^3 - 880x^5y^2 - 354x^5y - 8x^5$ $+ 2x^4y^8 + 22x^4y^7 + 106x^4y^6 + 452x^4y^5 + 834x^4y^4$ $+ 1454x^4y^3 + 770x^4y^2 - 140x^4y + 76x^4 - 16x^3y^8 - 68x^3y^7$ $- 384x^3y^6 - 722x^3y^5 - 1336x^3y^4 - 604x^3y^3 + 364x^3y^2 - 62x^3y$ $- 84x^3 + 4x^2y^9 + 34x^2y^8 + 120x^2y^7 + 446x^2y^6 + 308x^2y^5$ $+ 320x^2y^4 - 1946x^2y^3 - 1704x^2y^2 + 642x^2y - 1904x^2 + 4xy^{10}$ $+ 4xy^9 + 100xy^8 + 126xy^7 + 908xy^6 + 1328xy^5 + 3292xy^4 + 3470xy^3$ $+ 540xy^2 - 168xy + 620x - 2y^{11} - 22y^{10} - 118y^9 - 562y^8 - 1386y^7 - 3666y^6$ $- 4588y^5 - 6474y^4 - 2684y^3 + 1252y^2 - 534y + 11008$
B	$-16x^3y^{10} - 84x^3y^9 - 560x^3y^8 - 1656x^3y^7 - 5136x^3y^6$ $- 9196x^3y^5 - 15732x^3y^4 - 17728x^3y^3 - 16948x^3y^2 - 19372x^3y$ $- 19172x^3 + 4x^2y^{11} + 38x^2y^{10} + 186x^2y^9 + 840x^2y^8$ $+ 1964x^2y^7 + 4978x^2y^6 + 5286x^2y^5 + 6478x^2y^4 - 1586x^2y^3$ $- 8302x^2y^2 - 2210x^2y - 14444x^2 + 4xy^{12} + 8xy^{11} + 130xy^{10} + 254xy^9$ $+ 1628xy^8 + 3032xy^7 + 10274xy^6 + 17166xy^5 + 31542xy^4 + 42418xy^3$ $+ 40122xy^2 + 39826xy + 43180x - 2y^{13} - 24y^{12} - 154y^{11} - 846y^{10} - 2916y^9$ $- 9832y^8 - 22176y^7 - 49268y^6 - 75816y^5 - 108816y^4 - 118978y^3 - 97774y^2$ $- 102230y - 35648$

$i = 4, m = 1$, neighbours 43, 1, 2, 19, 28, 29, 34, 39

$W \quad 18x^3 + 54x^2 + 30x - 6$

$A \quad 3x^7y + 3x^7 + 3x^6y^2 + 15x^6y + 12x^6 + 21x^5y^2 + 39x^5y + 18x^5$
 $+ 9x^4y^3 + 57x^4y^2 + 63x^4y + 15x^4 - 3x^3y^4 + 33x^3y^3 + 66x^3y^2$
 $+ 33x^3y + 3x^3 - 3x^2y^5 - 9x^2y^4 + 27x^2y^3 - 27x^2y^2 - 78x^2y$
 $- 342x^2 - 21xy^5 - 27xy^4 - 120xy^3 - 264xy^2 - 168xy - 1962x - 3y^6 - 63y^5$
 $- 111y^4 - 255y^3 - 6y^2 + 1854y + 36$

$B \quad -6x^3y^6 - 12x^3y^4 - 150x^3y^3 - 447x^3y^2 - 1959x^3y - 4020x^3$
 $- 3x^2y^7 - 33x^2y^6 - 72x^2y^5 - 429x^2y^4 - 921x^2y^3$
 $- 2118x^2y^2 - 3933x^2y - 2673x^2 - 24xy^7 - 90xy^6 - 366xy^5 - 1050xy^4$
 $- 1527xy^3 - 3246xy^2 - 525xy + 1368x - 3y^8 - 63y^7 - 126y^6 - 618y^5 - 597y^4$
 $+ 429y^3 + 240y^2 + 11124y - 1638$

$i = 5, m = 1$, neighbours 44, 28, 39

$W \quad -3x^2 - 3x - 6$

$A \quad 6$

$B \quad -6x^2 - 12xy + 6y^2 - 15y - 9$

$i = 6, m = 1$, neighbours 42, 19, 22, 23, 26, 36

$W \quad -6x^2 + 6x + 16$

$A \quad 2x^6 - 8x^5 + 2x^4y + 8x^4 - 4x^3y + 4x^3 - 2x^2y^3 - 4x^2y^2 - 14x^2y$
 $- 4x^2 + 8xy^3 + 12xy^2 + 56xy - 12x - 8y^3 + 2y^2 - 28y - 6$

$B \quad 2x^3y^4 + 4x^3y^3 + 24x^3y^2 + 24x^3y + 22x^3 - 2x^2y^5 - 10x^2y^4$
 $- 30x^2y^3 - 78x^2y^2 - 36x^2y - 100x^2 + 8xy^5 + 12xy^4 + 88xy^3 + 44xy^2$
 $+ 180xy - 64x - 8y^5 + 8y^4 - 86y^3 + 102y^2 - 270y + 262$

$i = 7, m = 1$, neighbours 42, 19, 31, 39

$W \quad 2x^3 + 2x - 4$

$A \quad -2x^2 + 2x + 4y - 8$

$B \quad 2x^3y + 4x^3 - 2x^2y^2 + 2x^2y - 8x^2 - 2xy^2 + 6xy + 2x + 4y^3 - 2y^2 + 26y - 24$

$i = 8, m = 1$, neighbours 1, 12, 15, 24, 31

$W \quad -2x^3 + 5x^2 - 5x + 3$

$A \quad 2x^3 + x^2y - 3x^2 - 2xy^2 + xy + 15x - y^3 + 3y^2 - 12y - 47$

$B \quad -3x^3y^2 + 18x^3y - 31x^3 - x^2y^3 + 6x^2y^2 - 23x^2y + 37x^2 - 2xy^4$
 $+ 8xy^3 - 24xy^2 + 64xy - 47x - y^5 + 6y^4 - 33y^3 + 39y^2 - 150y + 28$

$i = 9, m = 1$, neighbours 12, 24, 36, 39

$W \quad 3x^2 - 2x$

$A \quad -2x^3 - x^2y + x^2 - 2xy + 2x + 2y^2 - 2$

$B \quad -3x^3y^2 + 2x^3y - 6x^3 - x^2y^3 + x^2y^2 - 3x^2y + 5x^2 - 3xy^2 + 4xy - x$
 $+ 2y^4 - 3y^3 + 13y^2 - 13y + 1$

$i = 10, m = 1$, neighbours 3, 21, 23, 28, 29, 36

$W \quad -165x^3 + 40x^2 - 255x + 20$

$A \quad -x^6y + x^6 - 3x^4y^2 + 5x^4y - 2x^4 + x^2y^4 - 3x^2y^3 + 11x^2y^2$
 $- 7x^2y - 27227x^2 - 2xy + 13202x - y^5 + 25y^4 - 437y^3 + 7197y^2 - 9104y + 400$

$B \quad 2x^3y^5 - 14x^3y^4 + 229x^3y^3 - 3616x^3y^2 + 4556x^3y - 43072x^3$
 $+ x^2y^6 - 7x^2y^5 + 114x^2y^4 - 1806x^2y^3 + 2272x^2y^2 - 21531x^2y$
 $+ 27112x^2 - 2xy^6 + 50xy^5 - 889xy^4 + 14712xy^3 - 10738xy^2 + 95236xy - 53529x$
 $- y^7 + 25y^6 - 444y^5 + 7346y^4 - 11749y^3 + 44023y^2 - 61822y + 11502$

$i = 11, m = 1$, neighbours 43, 1, 15

$W \quad 3x + 3$

$A \quad 0$

$B \quad 6x - 3y + 9$

$i = 12, m = 1$, neighbours 0, 8, 9, 15, 31, 37, 39

$W \quad -214x^3 + 324x^2 - 572x + 216$

$A \quad x^4y^2 + 2x^4y - 2x^3y^3 - 3x^3y^2 + 2x^3y + x^2y^4 + 2x^2y^3$
 $- 3x^2y^2 - 6x^2y - 45796x^2 - 2xy^4 - 3xy^3 + xy^2 - 2xy + 138672x - 8y^5$
 $+ 83y^4 - 988y^3 + 11959y^2 + 38510y - 57320$

$B \quad -4x^3y^5 + 41x^3y^4 - 490x^3y^3 + 5989x^3y^2 + 19254x^3y + 146236x^3$
 $+ x^2y^6 - 24x^2y^5 + 260x^2y^4 - 3248x^2y^3 - 6621x^2y^2 - 63486x^2y$
 $- 181332x^2 + 14xy^6 - 141xy^5 + 1748xy^4 - 21233xy^3 + 28416xy^2 - 78460xy$
 $+ 235144x - 8y^7 + 76y^6 - 955y^5 + 11515y^4 + 44001y^3 + 59389y^2 + 278462y - 103976$

$i = 13, m = 1$, neighbours 42, 17, 21, 24, 26, 35

$W \quad -2x^3 + 24x + 18$

A	$-4x^5 - 2x^4y + 12x^4 - 4x^3y - 4x^2y^2 + 20x^2y - 12x^2 + 4xy^3 + 4xy^2$
	$+12xy - 16x + 2y^4 - 10y^3 - 20y^2 + 4$
B	$2x^3y^3 - 22x^3y^2 + 44x^3y - 92x^3 - 6x^2y^4 + 14x^2y^3 - 24x^2y^2$
	$+28x^2y + 100x^2 + 2xy^5 + 14xy^4 - 8xy^3 + 96xy^2 - 196xy + 360x + 2y^6 - 14y^5$
	$+18y^4 - 88y^3 + 52y^2 - 70y + 160$
$i = 14, m = 1, \text{ neighbours } 2, 24, 32, 35$	
W	$-2x^3 + 4x^2 - 8x + 6$
A	$-x^2y - 5x^2 + 3xy + 19x + 2y^2 - 9y - 23$
B	$-x^3y^2 + 7x^3y - 28x^3 - x^2y^3 + x^2y^2 - 13x^2y + 41x^2 + 5xy^3$
	$-4xy^2 + 40xy - 55x + 2y^4 - 12y^3 + 11y^2 - 78y + 29$
$i = 15, m = 1, \text{ neighbours } 45, 2, 8, 11, 12, 22, 26, 32, 39$	
W	$324x^3 + 150x^2 - 1152x - 1518$
A	$24x^6y^2 + 72x^6y + 48x^6 + 24x^5y^3 + 48x^5y^2 - 24x^5y - 48x^5$
	$+6x^4y^4 - 6x^4y^3 - 114x^4y^2 - 210x^4y - 108x^4 - 6x^3y^4$
	$-138x^3y^3 - 342x^3y^2 - 150x^3y + 60x^3 - 24x^2y^5 - 204x^2y^4$
	$-396x^2y^3 - 60x^2y^2 + 276x^2y - 104856x^2 - 24xy^6 - 114xy^5 - 120xy^4$
	$+312xy^3 + 780xy^2 + 378xy - 97260x - 6y^7 + 54y^6 + 816y^5 + 3648y^4 + 6726y^3$
	$-23418y^2 - 141024y + 113532$
B	$-60x^3y^6 - 708x^3y^5 - 3180x^3y^4 - 5724x^3y^3 + 24516x^3y^2$
	$+141360x^3y + 610344x^3 - 54x^2y^7 - 444x^2y^6 - 1266x^2y^5$
	$+1848x^2y^4 + 37782x^2y^3 + 92688x^2y^2 + 327030x^2y + 213564x^2$
	$-30xy^8 - 132xy^7 + 282xy^6 + 5880xy^5 + 35814xy^4 + 121392xy^3 + 139362xy^2$
	$+188652xy - 1306440x - 6y^9 + 42y^8 + 858y^7 + 5682y^6 + 22950y^5 + 43590y^4$
	$-17154y^3 - 68046y^2 - 958812y - 1095396$
$i = 16, m = 1, \text{ neighbours } 42, 21, 26, 27, 28, 37$	
W	$-40x^3 - 56x^2 - 108x - 36$
A	$2x^6 - 4x^5 + 10x^4y + 8x^4 - 16x^3y - 10x^3 - 2x^2y^3 + 20x^2y^2 + 20x^2y$
	$-1594x^2 + 4xy^3 - 24xy^2 - 14xy - 4488x - 2y^4 - 20y^3 + 428y^2 + 1170y - 1792$
B	$-4x^3y^4 + 16x^3y^3 - 220x^3y^2 - 592x^3y - 4992x^3 - 2x^2y^5$
	$+18x^2y^4 - 120x^2y^3 - 164x^2y^2 - 2190x^2y - 4198x^2 - 68xy^4$
	$+728xy^3 - 1044xy^2 + 3784xy - 4304x - 2y^6 - 22y^5 + 392y^4 + 1436y^3 + 2226y^2$
	$+8826y - 3088$
$i = 17, m = 1, \text{ neighbours } 13, 23, 27, 29, 35, 37$	
W	$-x^3 + x^2 + 8x - 12$
A	$-x^7 - x^6y + x^6 + 5x^5 - 3x^4y^2 - x^4y - 6x^4 + x^3y^3 + 4x^3y^2$
	$+15x^3y + 2x^3 + x^2y^4 - 3x^2y^3 + 3x^2y^2 + x^2y + 13x^2 - 3xy^3$
	$+2xy^2 - 49xy - 12x + y^5 - 5y^4 + 6y^3 - 20y^2 + 49y - 18$
B	$-x^3y^5 - 4x^3y^4 - 11x^3y^3 - 16x^3y^2 + 7x^3y - 14x^3 + x^2y^6$
	$+3x^2y^5 + 14x^2y^4 + 14x^2y^3 + 31x^2y^2 + 8x^2y + 3x^2 - 2xy^6$
	$-xy^5 - 7xy^4 + 12xy^3 + 43xy^2 + 42xy + 20x + y^7 - 3y^6 + 5y^5 - 45y^4 - 2y^3$
	$-140y^2 + 41y - 79$
$i = 18, m = 1, \text{ neighbours } 23$	
W	$-x + 1$
A	0
B	1
$i = 19, m = 1, \text{ neighbours } 41, 2, 3, 4, 6, 7, 35$	
W	$x^5 - x^4 + x^3 - 5x^2 - x - 1$
A	$-x^6 + 2x^5 - 3x^4 - 2x^3y^2 + 3x^3y - 13x^3 + x^2y^3 + 6x^2y - 12x^2$
	$-2xy^3 - xy^2 - 12xy + 2x + y^4 - y^3 + 11y^2 - 16y + 1$
B	$2x^5y^2 - 3x^5y + 26x^5 - 2x^3y^3 + 3x^3y^2 - 26x^3y + 14x^3 - x^2y^4$
	$+3x^2y^3 - 7x^2y^2 + 32x^2y - 18x^2 + 2xy^5 + 22xy^3 + 7xy^2 + 60xy - 11x$
	$-y^6 + 2y^5 - 18y^4 + 21y^3 - 66y^2 + 108y - 23$
$i = 20, m = 1, \text{ neighbours } 3, 31, 34$	
W	$-x^2 + 1$
A	0
B	$-x^2y + y^2 + y - 1$
$i = 21, m = 1, \text{ neighbours } 46, 10, 13, 16, 35$	
W	$-x^5 - 6x^4 - 9x^3 - 6x^2 - 7x - 2$
A	$-x^6 - 12x^5 - 54x^4 + 11x^3y^2 - 9x^3y - 3x^3 + x^2y^3 + 6x^2y - 156x^2$
	$+12xy^3 - 11xy^2 + 81xy - 193x + y^4 - 12y^3 + 49y^2 - 213y + 4$

$$\begin{aligned}
B & -11x^5y^2 + 9x^5y - 116x^5 + 11x^4y^2 - 9x^4y + 116x^4 + 11x^3y^3 \\
& -9x^3y^2 + 116x^3y - 69x^3 - x^2y^4 + 44x^2y^3 - 42x^2y^2 + 314x^2y \\
& -168x^2 - 11xy^5 - xy^4 - 99xy^3 - 15xy^2 - 248xy + 161x - y^6 + 12y^5 - 55y^4 \\
& + 288y^3 - 310y^2 + 1321y - 233 \\
i = 22, m = 1, \text{ neighbours } 2, 3, 6, 15, 25, 29, 32 \\
W & -2x^5 + 3x^4 + 6x^3 + 3x^2 - 10x - 1 \\
A & -4x^6 + 12x^5 - x^4y^2 + 16x^4 - 2x^3y^3 + 8x^3y^2 + 2x^3y - 20x^3 \\
& + x^2y^4 + 4x^2y^3 - 5x^2y^2 + 24x^2y - 106x^2 + xy^4 - 8xy^3 - 76xy + 112x \\
& - y^5 - y^4 + 2y^3 - 54y^2 + 93y + 1 \\
B & 2x^5y^3 - 12x^5y^2 - 2x^5y + 4x^5 + 2x^4y^3 - 12x^4y^2 - 2x^4y + 4x^4 \\
& + 2x^3y^4 - 10x^3y^3 - 14x^3y^2 + 2x^3y - 92x^3 + x^2y^5 + 6x^2y^4 \\
& - 23x^2y^3 + 26x^2y^2 - 193x^2y + 62x^2 + 2xy^6 - 7xy^5 + 10xy^4 - 69xy^3 \\
& + 20xy^2 - 187xy + 224x - y^7 - 2y^6 - y^5 - 59y^4 + 100y^3 - 322y^2 + 615y - 89 \\
i = 23, m = 1, \text{ neighbours } 6, 10, 17, 18, 29 \\
W & -2x^5 + x^4 + 7x^3 - 4x^2 - 6x + 4 \\
A & -4x^6 + 4x^5 + 27x^4 + 4x^3y^2 - 8x^3y + 23x^3 + 4x^2y^3 + 24x^2y - 65x^2 \\
& - 4xy^3 - 20xy - 29x - 4y^4 - 41y^2 - 3y + 44 \\
B & -4x^5y^2 + 8x^5y - 49x^5 - 4x^3y^3 + 4x^3y^2 - 41x^3y + 41x^3 \\
& + 4x^2y^4 - 27x^2y^3 + 31x^2y^2 - 151x^2y - 4x^2 - 4xy^5 + 4xy^4 - 52xy^3 \\
& + 19xy^2 - 182xy + 33x - 4y^6 + 4y^5 - 70y^4 + 71y^3 - 268y^2 + 291y - 28 \\
i = 24, m = 1, \text{ neighbours } 1, 8, 9, 13, 14 \\
W & x^5 - x^4 + 3x^3 + 6x \\
A & -x^6 + 2x^5 - 7x^4 + 2x^3y^2 + 18x^3 + x^2y^3 + 6x^2y - 24x^2 - 2xy^3 - 12xy \\
& + 11x - y^4 + 2y^3 - 4y^2 + 14y - 20 \\
B & -2x^5y^2 - 11x^5 - 2x^3y^3 + 2x^3y^2 - 11x^3y + 21x^3 + x^2y^4 \\
& + 5x^2y^3 + 2x^2y^2 + 27x^2y - 43x^2 - 2xy^5 - 2xy^4 - 27xy^3 - 11xy^2 - 81xy \\
& + 23x - y^6 + y^5 - 8y^4 + 26y^3 - 2y^2 + 106y - 20 \\
i = 25, m = 1, \text{ neighbours } 46, 1, 22, 33 \\
W & -2x^2 - 4x \\
A & 0 \\
B & 2x^3 + 2x^2y - 8x^2 - 6xy - 12x - 2y^2 - 4y \\
i = 26, m = 1, \text{ neighbours } 0, 6, 13, 15, 16, 27, 28 \\
W & -4x^5 + 8x^4 + 12x^3 - 2x^2 - 16x - 6 \\
A & -16x^6 + 64x^5 - 2x^4y + 32x^4 + x^3y^3 - 41x^3y^2 - 90x^3y - 724x^3 \\
& + 19x^2y^3 - 7x^2y^2 + 80x^2y - 320x^2 + xy^4 - 62xy^3 - 23xy^2 - 406xy + 1444x \\
& + 20y^4 - 82y^3 - 24y^2 - 768y + 848 \\
B & -x^5y^3 + 40x^5y^2 + 81x^5y + 532x^5 - x^4y^3 + 40x^4y^2 + 81x^4y \\
& + 532x^4 + x^3y^4 - 42x^3y^3 - x^3y^2 - 370x^3y - 92x^3 - x^2y^5 \\
& - 11x^2y^4 - 36x^2y^3 - 8x^2y^2 - 190x^2y + 824x^2 - xy^6 + 42xy^5 \\
& + 129xy^4 + 867xy^3 + 817xy^2 + 3682xy + 408x - 21y^6 + 58y^5 - 35y^4 + 1260y^3 \\
& + 480y^2 + 5052y - 812 \\
i = 27, m = 1, \text{ neighbours } 3, 16, 17, 26, 32 \\
W & 2x^2 - 2x + 4 \\
A & -6x^3 + 4x^2y - 4x^2 - 10xy + 4x + 4y^2 - 10y - 6 \\
B & 6x^5 + 6x^4 - 6x^3y + 6x^3 - x^2y^3 + x^2y + 2x^2 + 7xy^3 - 2xy^2 + 41xy \\
& - 3y^4 + 7y^3 - 11y^2 + 59y - 24 \\
i = 28, m = 1, \text{ neighbours } 0, 4, 5, 10, 16, 26, 32 \\
W & -5x^5 - 6x^4 - 22x^3 - 10x^2 - 28x + 3 \\
A & -25x^6 - 60x^5 + 8x^4y - 256x^4 + 3x^3y^3 + 51x^3y^2 - 177x^3y + 451x^3 \\
& + 27x^2y^3 + 20x^2y^2 + 150x^2y - 849x^2 + 3xy^4 + 55xy^3 - 4xy^2 + 356xy \\
& + 1137x + 27y^4 + 30y^3 + 248y^2 + 148y - 867 \\
B & -3x^5y^3 - 63x^5y^2 + 165x^5y - 790x^5 + 3x^3y^4 + 69x^3y^3 \\
& - 39x^3y^2 + 460x^3y - 107x^3 - x^2y^5 - 40x^2y^4 + 219x^2y^3 \\
& - 335x^2y^2 + 1565x^2y - 1136x^2 - 3xy^6 - 52xy^5 + 97xy^4 - 909xy^3 \\
& + 731xy^2 - 3850xy + 1086x - 26y^6 + 19y^5 - 445y^4 + 588y^3 - 1767y^2 + 2894y - 438 \\
i = 29, m = 1, \text{ neighbours } 42, 3, 4, 10, 17, 22, 23, 38 \\
W & 4x^5 + 4x^4 - 10x^3 - 4x^2 - 6x + 6 \\
A & -16x^6 + 8x^5y - 40x^5 - 8x^4y^2 + 8x^4y + 64x^4 - 6x^3y^3 + 12x^3y^2 \\
& - 40x^3y + 380x^3 + 2x^2y^4 + 16x^2y^3 + 18x^2y^2 + 100x^2y - 28x^2 \\
& - 4xy^4 + 56xy^3 - 54xy^2 + 252xy - 360x + 2y^5 + 28y^4 - 52y^3 + 44y^2 - 168y + 140
\end{aligned}$$

B	$4x^5y^3 - 14x^5y^2 + 4x^5y - 212x^5 - 4x^4y^3 + 14x^4y^2 - 4x^4y$ $+ 212x^4 - 4x^3y^4 + 18x^3y^3 - 18x^3y^2 + 216x^3y - 272x^3 + 4x^2y^5$ $- 14x^2y^4 - 38x^2y^3 - 84x^2y^2 - 592x^2y - 124x^2 + 6xy^6 - 18xy^5$ $+ 52xy^4 - 378xy^3 + 102xy^2 - 1352xy + 324x - 2y^7 - 28y^6 + 6y^5 - 202y^4$ $+ 312y^3 - 148y^2 + 904y - 104$
$i = 30, m = 1$, neighbours 39	
W	-1
A	0
B	1
$i = 31, m = 1$, neighbours 41, 7, 8, 12, 20, 35, 39	
W	$-3x^5 + 10x^3 + 13x^2 - 16x - 16$
A	$-9x^6 + 2x^4y + 64x^4 - 13x^3y^2 + 60x^3y - 179x^3 + 10x^2y^3 + 2x^2y^2$ $+ 47x^2y - 219x^2 + 3xy^3 - 9xy^2 - 28xy + 449x - 10y^4 + 8y^3 - 101y^2 + 166y + 317$
B	$16x^5y^2 - 57x^5y + 260x^5 + 16x^4y^2 - 57x^4y + 260x^4 + 16x^3y^3$ $- 25x^3y^2 + 146x^3y - 341x^3 + 9x^2y^4 - 46x^2y^3 + 25x^2y^2 - 243x^2y$ $+ 8x^2 + 13xy^5 - 38xy^4 + 197xy^3 - 206xy^2 + 650xy + 441x - 10y^6 + 19y^5$ $- 180y^4 + 400y^3 - 695y^2 + 1795y - 61$
$i = 32, m = 1$, neighbours 1, 2, 14, 15, 22, 27, 28	
W	$-x^5 - 3x^4 + 7x^3 - 13x^2 + 8x - 6$
A	$-x^6 + x^5y - 5x^5 - x^4y^2 - 3x^4y + 3x^4 + 3x^3y^3 - x^3y^2 + 21x^3y$ $- 32x^3 + x^2y^4 + 3x^2y^3 - 9x^2y^2 - 2x^2y - 104x^2 - 4xy^4 - xy^3$ $+ 2xy^2 + 32xy + 105x - y^5 - 4y^4 + 13y^3 - 19y^2 + 141y - 130$
B	$-4x^5y^3 + 3x^5y^2 - 18x^5y + 49x^5 - 4x^4y^3 + 3x^4y^2 - 18x^4y$ $+ 49x^4 - 4x^3y^4 + 3x^3y^3 - 18x^3y^2 + 49x^3y + 93x^3 + 3x^2y^5$ $+ 7x^2y^4 + 4x^2y^3 - 15x^2y^2 - 62x^2y - 90x^2 - 3xy^6 - xy^5 - 34xy^4$ $+ 20xy^3 - 44xy^2 + 197xy + 165x - y^7 - 4y^6 + 7y^5 - 34y^4 + 175y^3 - 102y^2 + 755y$ $- 166$
$i = 33, m = 1$, neighbours 1, 2, 25, 34	
W	$x^3 + 2x^2 - x - 2$
A	$-x^3 - x^2 + xy + 2x + y + 2$
B	$x^5 + x^3y - 3x^3 + x^2y^2 + 2x^2y - x^2 + xy^3 + xy^2 + 5xy + 2x + 5y + 1$
$i = 34, m = 1$, neighbours 46, 3, 4, 20, 33, 36	
W	$-x^5 - x^4 + 5x^3 + x^2 - 8x + 4$
A	$-x^6 - 2x^5 + 9x^4 - x^3y^2 - 4x^3y + 19x^3 + x^2y^3 - 2x^2y^2 + 4x^2y$ $- 34x^2 + 2xy^3 + 3xy^2 + 14xy - 46x - y^4 + y^3 - 15y^2 + 17y + 36$
B	$x^5y^2 + 6x^5y - 8x^5 - x^4y^2 - 6x^4y + 8x^4 + x^3y^3 + 7x^3y^2$ $- 2x^3y + 3x^3 + x^2y^4 - 10x^2y^3 + 10x^2y^2 - 65x^2y + 38x^2 + xy^5$ $+ 4xy^4 - 6xy^3 + 18xy^2 - 48xy - 2x - y^6 + 2y^5 - 21y^4 + 49y^3 - 106y^2 + 219y - 20$
$i = 35, m = 32$, neighbours 41, 3, 13, 14, 17, 19, 21, 31, 38	
W	$-512x^5 + 2368x^4 - 1408x^3 - 256x^2 - 3264x - 1152$
A	$2048x^7y - 2048x^7 + 1024x^6y^2 + 1024x^6y - 264192x^6 - 2048x^5y^3$ $- 16384x^5y + 2443264x^5 - 1024x^4y^4 - 3072x^4y^3 - 15360x^4y^2$ $+ 5120x^4y - 7034880x^4 + 2048x^3y^5 + 2560x^3y^4 + 71808x^3y^3$ $- 1608288x^3y^2 - 4338104x^3y - 20779414x^3 - 3072x^2y^5 + 263168x^2y^3$ $+ 46080x^2y^2 + 1567744x^2y - 6838272x^2 + 2048xy^6 - 3584xy^5 + 26240xy^4$ $- 2510048xy^3 + 2906664xy^2 - 7080190xy + 149231598x + 2048y^6 + 10240y^5$ $+ 368896y^4 - 3984832y^3 - 2702000y^2 - 55743420y + 61827884$
B	$-2048x^5y^5 - 10752x^5y^4 - 77952x^5y^3 + 1600096x^5y^2 + 4411832x^5y$ $+ 27396502x^5 - 4096x^4y^5 - 21504x^4y^4 - 155904x^4y^3 + 3200192x^4y^2$ $+ 8823664x^4y + 54793004x^4 + 2048x^3y^6 + 4096x^3y^5 + 43008x^3y^4$ $- 1853440x^3y^3 + 788480x^3y^2 - 13058048x^3y + 10829824x^3 + 1024x^2y^7$ $+ 9216x^2y^6 + 55296x^2y^5 - 88576x^2y^4 + 3971968x^2y^3$ $+ 108128x^2y^2 + 30243256x^2y + 24576406x^2 - 2048xy^8 - 2560xy^7 - 57472xy^6$ $+ 1702496xy^5 + 5651640xy^4 + 35452758xy^3 + 38403248xy^2 + 155432764xy + 41118102x$ $- 2048y^8 - 11264y^7 - 391424y^6 + 3526848y^5 + 4020592y^4 + 84056236y^3$ $+ 34459616y^2 + 366209944y - 60500780$
$i = 36, m = 2$, neighbours 41, 2, 3, 6, 9, 10, 34	
W	$4x^4 - 10x^3 + 4x^2$
A	$-16x^4 - 20x^3y + 31x^3 - 132x^2 + 8xy^2 + 35xy + 215x - 8y^3 - 24y^2 - 118y - 102$
B	$20x^5y + 49x^5 + 40x^4y + 98x^4 - 20x^3y^2 - 24x^3y + 20x^3 + 20x^2y^3$ $- 48x^2y^2 + 52x^2y - 27x^2 + 12xy^4 + xy^3 + 60xy^2 - 70xy + 105x + 4y^5 + 8y^4$

	$+150y^3 + 96y^2 + 600y - 102$
$i = 37, m = 1$, neighbours 0, 12, 16, 17	
W	$x^2 - 3x + 3$
A	$2x + y - 1$
B	$-2x^3 - x^2y - 2x^2 - 7xy + 8x - 3y^2 + 5y - 5$
$i = 38, m = 1$, neighbours 29, 35	
W	$-2x$
A	0
B	$x^2 - 3x + y - 1$
$i = 39, m = 32$, neighbours 2, 4, 5, 7, 9, 12, 15, 30, 31	
W	$32x^5 + 160x^4 + 480x^3 + 576x^2 + 832x + 512$
A	$4096x^6y + 3072x^6 - 4096x^5y^2 - 2048x^5y - 8192x^5 - 7168x^4y^3$ $-19456x^4y - 82944x^4 - 2048x^3y^5 + 512x^3y^4 + 10112x^3y^3$ $+31392x^3y^2 + 9720x^3y - 59526x^3 + 4096x^2y^5 - 4096x^2y^4$ $+31744x^2y^3 - 14336x^2y^2 + 6144x^2y - 405504x^2 - 2048xy^6 + 14848xy^5$ $+25984xy^4 + 14368xy^3 - 33256xy^2 + 61842xy - 842350x + 8192y^6 + 8192y^5$ $-18176y^4 - 74816y^3 - 86320y^2 - 502500y - 440052$
B	$2048x^5y^5 + 5632x^5y^4 - 17280x^5y^3 - 41632x^5y^2 - 10744x^5y$ $-134010x^5 - 4096x^4y^5 - 11264x^4y^4 + 34560x^4y^3 + 83264x^4y^2$ $+21488x^4y + 268020x^4 - 2048x^3y^6 - 4096x^3y^5 + 21504x^3y^4$ $+28672x^3y^3 - 20480x^3y^2 + 125952x^3y - 359424x^3 - 1024x^2y^7$ $+7168x^2y^6 - 2048x^2y^5 + 29184x^2y^4 - 15232x^2y^3 + 82272x^2y^2$ $+403976x^2y - 443258x^2 + 2048xy^8 + 2560xy^7 - 16256xy^6 - 53920xy^5$ $-197880xy^4 - 161978xy^3 - 289776xy^2 - 211700xy + 150126x - 7168y^8 - 2048y^7$ $-33024y^6 + 68928y^5 + 139760y^4 + 792436y^3 + 88672y^2 + 2840008y - 702196$

Table 2: The data of the identities (3).

$\{i, j\} = \{0, 12\}, k_{ij} = 7$	
α	-16
β	$8x^3 + 4x^2 + 2x - 16y^2 - 87$
γ	$-8x^3 - 4x^2 - 2x - 9$
$\{i, j\} = \{0, 26\}, k_{ij} = 11$	
α	$2x^3 - 7$
β	$-2x^5 - 2x^4 + 2x^3y^2 + 8x^3 - x^2 - 3x - 7y^2 - 46$
γ	$-2x^5 - 2x^4 - 4x^3 - x^2 - 3x - 4$
$\{i, j\} = \{0, 28\}, k_{ij} = 2 \cdot 11$	
α	$x^3 + 6$
β	$-x^5 + x^3y^2 + 8x^3 - 7x^2 - 3x + 6y^2 + 50$
γ	$-x^5 + 2x^3 - 7x^2 - 3x + 14$
$\{i, j\} = \{0, 37\}, k_{ij} = 7$	
α	$-2x^3 - 5$
β	$2x^5 + 2x^4 - 2x^3y^2 - 14x^3 + x^2 + 3x - 5y^2 - 29$
γ	$2x^5 + 2x^4 - 2x^3 + x^2 + 3x + 1$
$\{i, j\} = \{1, 4\}, k_{ij} = 2$	
α	-2
β	$-2x^3 + 2x^2y - 2xy^2 + y^3 - y^2 - 7$
γ	$-y^3 - y^2 - 5$
$\{i, j\} = \{1, 8\}, k_{ij} = 3$	
α	-1
β	$-x^3 + x^2y - xy^2 + y^3 - 1$
γ	$-y^3 + y^2 - 2y - 1$
$\{i, j\} = \{1, 11\}, k_{ij} = 2 \cdot 3$	
α	1
β	$x^3 - x^2y + xy^2 + y^3 + 4y^2 + 4y + 15$
γ	$-y^3 - 2y^2 - 2y - 7$
$\{i, j\} = \{1, 24\}, k_{ij} = 2 \cdot 3 \cdot 5$	
α	$2x^4 - 8y + 5$

β	$2x^7 - 2x^6y - 4x^5y - 2x^5 + 4x^4y^2 + 2x^4y + 2x^4 + x^3 - x^2y + 8xy^2$
	$+2xy + 35x - 8y^3 - 2y^2 - 43y + 5$
γ	$-2x^4y^2 - 4x^4y - 2x^4 + 7y^2 + 2y + 35$
$\{i, j\} = \{1, 25\}, k_{ij} = 3^3 \cdot 11$	
α	$-17x^4 + 289y - 297$
β	$-17x^7 + 17x^6y + 51x^5y + 102x^5 - 34x^4y^2 - 51x^4y + 85x^4 - 280x^3$
	$+280x^2y - 289xy^2 + 279xy - 887x + 289y^3 - 288y^2 + 1455y - 1184$
γ	$17x^4y^2 + 51x^4y + 102x^4 - 9y^2 + 279y - 887$
$\{i, j\} = \{1, 32\}, k_{ij} = 2^4$	
α	$x^4 - 4y - 1$
β	$x^7 - x^6y - x^5y + x^5 + 2x^4y^2 - x^3 + x^2y + 4xy^2 + xy + 15x - 4y^3 - 6y^2$
	$-20y - 16$
γ	$-x^4y^2 - x^4y + x^4 + 5y^2 + y + 15$
$\{i, j\} = \{1, 33\}, k_{ij} = 2^4 \cdot 3^2 \cdot 11$	
α	$-13x^4 + 169y - 82$
β	$-13x^7 + 13x^6y + 26x^5y + 52x^5 - 26x^4y^2 - 52x^4y - 13x^4 - 173x^3$
	$+173x^2y - 169xy^2 + 86xy - 751x + 169y^3 - 86y^2 + 920y - 82$
γ	$13x^4y^2 + 26x^4y + 52x^4 + 4y^2 + 86y - 751$
$\{i, j\} = \{2, 3\}, k_{ij} = 2$	
α	0
β	-1
γ	1
$\{i, j\} = \{2, 4\}, k_{ij} = 2^3$	
α	-1
β	$-x^3 + x^2 - x - y^2 + y - 7$
γ	$x^3 - x^2 + x$
$\{i, j\} = \{2, 14\}, k_{ij} = 2 \cdot 3$	
α	-1
β	$-x^3 - x^2 - x - y^2 + y - 9$
γ	$x^3 + x^2 + x + 2$
$\{i, j\} = \{2, 15\}, k_{ij} = 2 \cdot 5 \cdot 7$	
α	1
β	$x^3 - 3x^2 + 9x + y^2 - y - 19$
γ	$-x^3 + 3x^2 - 9x + 26$
$\{i, j\} = \{2, 19\}, k_{ij} = 2^4 \cdot 3$	
α	$-x^3 - 7$
β	$x^5 - x^3y^2 + x^3y - 6x^3 + 8x^2 - 7x - 7y^2 + 7y - 41$
γ	$x^5 + x^3 + 8x^2 - 7x + 8$
$\{i, j\} = \{2, 22\}, k_{ij} = 2^3$	
α	-1
β	$-x^2 - x - y^2 + y - 7$
γ	$x^2 + x$
$\{i, j\} = \{2, 32\}, k_{ij} = 2^2 \cdot 7$	
α	$-x^3 - 6$
β	$-x^5 - x^4 - x^3y^2 + x^3y - 6x^3 - 4x^2 + 2x - 6y^2 + 6y - 32$
γ	$x^5 + x^4 - x^3 + 4x^2 - 2x - 10$
$\{i, j\} = \{2, 33\}, k_{ij} = 2^4 \cdot 3$	
α	$-x^3 - 7$
β	$-x^5 - x^3y^2 + x^3y - 8x^3 - 8x^2 + 7x - 7y^2 + 7y - 57$
γ	$x^5 + x^3 + 8x^2 - 7x + 8$
$\{i, j\} = \{2, 36\}, k_{ij} = 2^6 \cdot 5$	
α	$9x^3 - 40$
β	$-9x^5 - 18x^4 + 9x^3y^2 - 9x^3y + 27x^3 - 41x^2 - 10x - 40y^2 + 40y - 300$
γ	$-9x^5 - 18x^4 - 36x^3 - 41x^2 - 10x - 20$
$\{i, j\} = \{2, 39\}, k_{ij} = 2^6 \cdot 3$	
α	$-7x^3 - 24$
β	$7x^5 - 14x^4 - 7x^3y^2 + 7x^3y - 21x^3 - 25x^2 - 6x - 24y^2 + 24y - 156$
γ	$7x^5 - 14x^4 + 28x^3 - 25x^2 - 6x + 12$
$\{i, j\} = \{3, 10\}, k_{ij} = 3 \cdot 5 \cdot 7$	

α	16
β	$8x^3 + 4x^2 + 2x + 16y^2 + 16y + 121$
γ	$-8x^3 - 4x^2 - 2x - 9$
$\{i, j\}$	$\{3, 19\}, k_{ij} = 2 \cdot 5^2$
α	$x^3 + 7$
β	$-x^5 + x^3y^2 + x^3y + 8x^3 - 8x^2 - 7x + 7y^2 + 7y + 57$
γ	$-x^5 + x^3 - 8x^2 - 7x + 8$
$\{i, j\}$	$\{3, 20\}, k_{ij} = 2^4 \cdot 3$
α	$-x^3 + 7$
β	$-x^5 - x^3y^2 - x^3y - 8x^3 + 6x^2 - 7x + 7y^2 + 7y + 55$
γ	$x^5 + x^3 - 6x^2 + 7x - 6$
$\{i, j\}$	$\{3, 22\}, k_{ij} = 2^3 \cdot 5$
α	$-x^3 + 7$
β	$-x^5 - x^4 - x^3y^2 - x^3y - 9x^3 + 3x^2 - 5x + 7y^2 + 7y + 47$
γ	$x^5 + x^4 + 2x^3 - 3x^2 + 5x + 2$
$\{i, j\}$	$\{3, 27\}, k_{ij} = 2^2 \cdot 7$
α	$-x^3 + 1$
β	$x^5 + x^4 - x^3y^2 - x^3y - 8x^3 - 3x^2 + 5x + y^2 + y + 18$
γ	$x^5 + x^4 - x^3 - 3x^2 + 5x + 11$
$\{i, j\}$	$\{3, 29\}, k_{ij} = 2 \cdot 3$
α	1
β	$-x^2 + x + y^2 + y + 6$
γ	$-x^2 + x - 1$
$\{i, j\}$	$\{3, 34\}, k_{ij} = 2^3 \cdot 3 \cdot 5$
α	$x^3 + 14$
β	$x^5 - x^4 + x^3y^2 + x^3y + 10x^3 + 10x^2 + 2x + 14y^2 + 14y + 116$
γ	$-x^5 + x^4 - 3x^3 - 10x^2 - 2x - 18$
$\{i, j\}$	$\{3, 35\}, k_{ij} = 2^6 \cdot 3$
α	$-3x^3 + 32$
β	$3x^5 + 6x^4 - 3x^3y^2 - 3x^3y - 9x^3 - 5x^2 + 8x + 32y^2 + 32y + 240$
γ	$3x^5 + 6x^4 + 12x^3 - 5x^2 + 8x + 16$
$\{i, j\}$	$\{3, 36\}, k_{ij} = 2^2 \cdot 5$
α	$-x^3 + 4$
β	$x^5 + 2x^4 - x^3y^2 - x^3y - 5x^3 - 3x^2 - 4x + 4y^2 + 4y + 26$
γ	$x^5 + 2x^4 + 2x^3 - 3x^2 - 4x - 2$
$\{i, j\}$	$\{4, 19\}, k_{ij} = 2^6$
α	$7x^4 - 49y - 64$
β	$-7x^7 - 7x^6y - 7x^4 + 71x^3 + 71x^2y - 49xy^2 + 56xy - 399x - 49y^3 + 56y^2 - 350y$
γ	$+64$ $-7x^4y^2 + 120y^2 - 56y + 399$
$\{i, j\}$	$\{4, 28\}, k_{ij} = 2^5 \cdot 3$
α	$x^4 - 9y - 6$
β	$-x^7 - x^6y - 2x^5 - 2x^4y - x^4 + 3x^3 + 3x^2y - 9xy^2 + 6xy - 45x - 9y^3 + 6y^2$
γ	$-36y + 6$ $-x^4y^2 + 2x^4 + 12y^2 - 6y + 45$
$\{i, j\}$	$\{4, 29\}, k_{ij} = 2$
α	1
β	$-x^3 - x^2y - xy + 3x + 2y + 2$
γ	$-y^2 + y - 3$
$\{i, j\}$	$\{4, 34\}, k_{ij} = 2^6$
α	$-x^4 + 8y + 1$
β	$x^7 + x^6y - x^5y - x^5 - 2x^4y^2 - 2x^4y - x^3 - x^2y + 8xy^2 + xy + 65x + 8y^3$
γ	$+10y^2 + 58y + 64$ $x^4y^2 + x^4y + x^4 - 9y^2 - y - 65$
$\{i, j\}$	$\{4, 39\}, k_{ij} = 2 \cdot 3^2 \cdot 5$
α	$x^4 + 6y - 22$
β	$-x^7 - x^6y - 2x^5y + 5x^5 + x^4y + 9x^4 + 28x^3 + 28x^2y + 6xy^2 + 44xy - 68x$
γ	$+6y^3 + 14y - 114$ $-x^4y^2 + 2x^4y - 5x^4 + 22y^2 - 44y + 68$

$$\begin{aligned}
&\{i, j\} = \{5, 28\}, k_{ij} = 2^2 \cdot 3^6 \\
&\alpha \quad 4x^4 - 54y - 92 \\
&\beta \quad 2x^7 - x^6y - 6x^5y + 28x^5 + 3x^4y^2 - 14x^4y + 2x^4 + 8x^3 - 4x^2y \\
&\quad + 108xy^2 - 240xy + 706x - 54y^3 + 120y^2 - 380y - 46 \\
&\gamma \quad -x^4y^2 - 12x^4y + 56x^4 + 212y^2 - 480y + 1412 \\
&\{i, j\} = \{5, 39\}, k_{ij} = 2^2 \cdot 3^2 \\
&\alpha \quad 4x^4 - 6y - 4 \\
&\beta \quad 2x^7 - x^6y - 8x^5y + 2x^5 + 3x^4y^2 - 17x^4y + 6x^4 - 2x^3 + x^2y + 12xy^2 \\
&\quad + 8xy + 16x - 6y^3 + 21y^2 + 5y + 30 \\
&\gamma \quad -x^4y^2 - 16x^4y + 4x^4 + 25y^2 + 16y + 32 \\
&\{i, j\} = \{6, 19\}, k_{ij} = 2 \cdot 5 \\
&\alpha \quad x^4 + 2y - 2 \\
&\beta \quad x^7 + x^4y^2 + x^4y + x^4 + 4x^3 + 6x + 4y^2 - 8 \\
&\gamma \quad x^5y + x^5 + 2x^2y^2 - 6x^2y + 8x^2 + 4xy + 4x - 2y^3 + 6y^2 - 8y + 6 \\
&\{i, j\} = \{6, 22\}, k_{ij} = 5 \cdot 13 \\
&\alpha \quad x^4 - 7y - 12 \\
&\beta \quad x^7 - x^6y - x^6 - x^5y + 3x^5 + 2x^4y^2 + x^4y - 4x^4 - 9x^3 + 9x^2y + 9x^2 \\
&\quad + 7xy^2 - 2xy + 16x - 7y^3 - 21y^2 - 37y - 53 \\
&\gamma \quad -x^4y^2 - 3x^4y + 2x^4 + 16y^2 + 16y + 25 \\
&\{i, j\} = \{6, 23\}, k_{ij} = 2^4 \cdot 3 \\
&\alpha \quad 7x^4 - 49y - 55 \\
&\beta \quad 7x^7 - 7x^6y - 7x^6 - 14x^5y - 7x^5 + 14x^4y^2 + 21x^4y + 14x^4 - 62x^3 \\
&\quad + 62x^2y + 62x^2 + 49xy^2 + 68xy + 363x - 49y^3 - 117y^2 - 480y - 418 \\
&\gamma \quad -7x^4y^2 - 28x^4y - 14x^4 + 111y^2 + 192y + 425 \\
&\{i, j\} = \{6, 26\}, k_{ij} = 2^2 \cdot 5 \cdot 13 \\
&\alpha \quad x^4 - 4y - 86 \\
&\beta \quad x^7 - x^6y - x^6 + x^5y + 2x^4 - 88x^3 + 88x^2y + 88x^2 + 4xy^2 - 106xy + 88x \\
&\quad - 4y^3 + 10y^2 - 56y - 350 \\
&\gamma \quad -x^4y^2 - x^4y - x^4 + 92y^2 + 70y + 176 \\
&\{i, j\} = \{6, 36\}, k_{ij} = 2^4 \cdot 5 \cdot 13 \\
&\alpha \quad -x^4 - y - 131 \\
&\beta \quad -x^7 + x^6y + x^6 - 2x^5y + x^5 + x^4y - 6x^4 - 122x^3 + 122x^2y + 122x^2 \\
&\quad + xy^2 - 240xy + 103x - y^3 - 7y^2 + 128y - 684 \\
&\gamma \quad x^4y^2 + 2x^4 + 123y^2 + 4y + 225 \\
&\{i, j\} = \{7, 19\}, k_{ij} = 2 \cdot 5 \cdot 7 \\
&\alpha \quad 3x^4 - 18y + 13 \\
&\beta \quad 3x^7 - 3x^6y + 3x^6 - 6x^5 + 6x^4y - 3x^4 + 7x^3 - 7x^2y + 7x^2 + 18xy^2 + 12xy \\
&\quad + 70x - 18y^3 + 6y^2 - 76y + 83 \\
&\gamma \quad -3x^4y^2 + 6x^4y - 9x^4 + 11y^2 + 26y + 63 \\
&\{i, j\} = \{7, 31\}, k_{ij} = 2^3 \cdot 7 \\
&\alpha \quad x^4 - 10y + 22 \\
&\beta \quad x^7 - x^6y + x^6 - x^5y - 4x^5 + 2x^4y^2 + 2x^4y + 2x^4 + 28x^3 - 28x^2y \\
&\quad + 28x^2 + 10xy^2 - 2xy - 10y^3 + 30y^2 - 66y + 50 \\
&\gamma \quad -x^4y^2 + x^4y - 5x^4 - 18y^2 + 54y - 28 \\
&\{i, j\} = \{7, 39\}, k_{ij} = 2^2 \cdot 5 \cdot 7 \\
&\alpha \quad x^4 - 7y + 15 \\
&\beta \quad x^7 - x^6y + x^6 - 2x^5y - x^5 - x^4y - 4x^4 + 8x^3 - 8x^2y + 8x^2 + 7xy^2 - 2xy \\
&\quad + 41x - 7y^3 + 7y^2 - 22y + 122 \\
&\gamma \quad -x^4y^2 - 2x^4 - y^2 + 14y + 33 \\
&\{i, j\} = \{8, 12\}, k_{ij} = 3 \cdot 7 \cdot 11 \\
&\alpha \quad -16 \\
&\beta \quad 8x^3 + 12x^2 + 18x - 16y^2 + 32y - 125 \\
&\gamma \quad -8x^3 - 12x^2 - 18x - 35 \\
&\{i, j\} = \{8, 15\}, k_{ij} = 3 \cdot 7 \cdot 11 \\
&\alpha \quad 1 \\
&\beta \quad x^3 - 4x^2 + 16x + y^2 - 2y - 53 \\
&\gamma \quad -x^3 + 4x^2 - 16x + 63 \\
&\{i, j\} = \{8, 24\}, k_{ij} = 3^2 \cdot 7^2 \\
&\alpha \quad x^3 - 36
\end{aligned}$$

β	$x^5 + x^3y^2 - 2x^3y + 7x^3 - 35x^2 - 12x - 36y^2 + 72y - 255$
γ	$-x^5 + 3x^3 + 35x^2 + 12x - 105$
$\{i, j\}$	$\{8, 31\}, k_{ij} = 3 \cdot 7$
α	-1
β	$-x^2 - x - y^2 + 2y - 10$
γ	$x^2 + x$
$\{i, j\}$	$\{9, 12\}, k_{ij} = 2^4 \cdot 11$
α	-81
β	$81x^3 + 81x^2y + 81x^2 + 81xy^2 + 162xy + 81x + 27y^3 - 9y^2 + 3y - 352$
γ	$-27y^3 - 126y^2 - 120y - 257$
$\{i, j\}$	$\{9, 24\}, k_{ij} = 2^5$
α	$x^4 - 3y - 3$
β	$-x^7 - x^6y - x^6 + 2x^5y - x^5 + 2x^4y^2 + x^4y - 2x^4 + 2x^3 + 2x^2y$ $+ 2x^2 - 3xy^2 - 2xy - 15x - 3y^3 - 5y^2 - 14y - 12$
γ	$-x^4y^2 - 4x^4y + 5y^2 + 6y + 17$
$\{i, j\}$	$\{9, 36\}, k_{ij} = 2^6$
α	$-9x^4 + 81y + 73$
β	$9x^7 + 9x^6y + 9x^6 - 18x^5y - 9x^5 + 45x^4y + 36x^4 - 64x^3 - 64x^2y - 64x^2$ $+ 81xy^2 + 56xy + 703x + 81y^3 - 153y^2 + 310y - 904$
γ	$9x^4y^2 + 36x^4y + 18x^4 - 145y^2 - 184y - 767$
$\{i, j\}$	$\{9, 39\}, k_{ij} = 2^5 \cdot 7 \cdot 11$
α	$x^4 - y - 309$
β	$-x^7 - x^6y - x^6 - 2x^5y + x^5 - x^4y + 4x^4 + 316x^3 + 316x^2y + 316x^2$ $- xy^2 + 644xy - 379x - y^3 + 9y^2 + 290y - 1460$
γ	$-x^4y^2 - 2x^4 + 317y^2 - 12y + 695$
$\{i, j\}$	$\{10, 21\}, k_{ij} = 5^3 \cdot 7$
α	$1102x^4 + 14732y - 875$
β	$-2552x^7 + 58x^6y - 290x^5y + 1740x^5 + 1160x^4y^2 + 3248x^4y - 812x^4$ $- 11834x^3 - 14933x^2y + 13456xy^2 - 6535xy + 6016x + 10730y^3 - 16040y^2 + 35107y$ $- 6369$
γ	$4002x^6 - 2668x^5y - 4002x^5 + 58x^4y^2 - 754x^4y + 522x^4 + 4002x^2y^2$ $- 22678x^2y + 25645x^2 - 4002xy^2 + 40710xy - 25645x - 4002y^3 - 15165y^2 - 53285y$ $+ 13613$
$\{i, j\}$	$\{10, 23\}, k_{ij} = 2^2 \cdot 5 \cdot 7^2$
α	$-728x^4 + 4394y + 3520$
β	$-1326x^7 + 13x^6y - 130x^5y - 546x^5 - 741x^4y^2 - 949x^4y + 364x^4$ $- 4906x^3 + 1785x^2y + 3380xy^2 - 2250xy + 7268x + 1690y^3 - 5363y^2 - 2801y - 1760$
γ	$3380x^6 - 1352x^5y + 13x^4y^2 + 3640x^4y + 4472x^4 + 2704x^2y^2 + 8450x^2y$ $+ 10036x^2 - 8476xy + 2704y^3 + 6179y^2 + 22986y - 4500$
$\{i, j\}$	$\{10, 28\}, k_{ij} = 2^2 \cdot 5 \cdot 13^2$
α	$4x^4 - 58y - 108$
β	$-2x^7 - x^6y + 6x^5y - 28x^5 + 3x^4y^2 - 14x^4y - 2x^4 - 116xy^2 + 248xy - 818x$ $- 58y^3 + 124y^2 - 380y + 54$
γ	$-x^4y^2 - 12x^4y + 56x^4 + 232y^2 - 496y + 1636$
$\{i, j\}$	$\{10, 29\}, k_{ij} = 3^2 \cdot 5$
α	$348x^4 + 5744y - 3467$
β	$-888x^7 + 32x^6y - 160x^5y + 896x^5 + 380x^4y^2 + 764x^4y - 1420x^4$ $+ 15470x^3 - 10472x^2y + 7680xy^2 + 2440xy - 27772x + 4792y^3 - 15971y^2 + 11743y$ $+ 13053$
γ	$1428x^6 - 952x^5y - 1428x^5 + 32x^4y^2 - 156x^4y + 1064x^4 + 952x^2y^2$ $- 10472x^2y - 26061x^2 - 952xy^2 + 46886xy + 26061x - 952y^3 - 13456y^2 - 36177y + 3422$
$\{i, j\}$	$\{10, 36\}, k_{ij} = 2^2 \cdot 5^2$
α	$4x^4 - 10y - 4$
β	$-2x^7 - x^6y + 8x^5y - 2x^5 + 3x^4y^2 - 17x^4y + 2x^4 + 2x^3 + x^2y - 20xy^2$ $- 8xy - 48x - 10y^3 + 37y^2 - 3y + 98$
γ	$-x^4y^2 - 16x^4y + 4x^4 + 41y^2 + 16y + 96$
$\{i, j\}$	$\{11, 15\}, k_{ij} = 2^4 \cdot 3^2$
α	24
β	$12x^3 - 6x^2y - 6x^2 + 3xy^2 + 6xy + 3x - y^3 + 6y^2 + 9y + 70$
γ	$y^3 + 21y^2 + 27y + 119$

$$\begin{aligned}
&\{i, j\} = \{12, 15\}, k_{ij} = 2^4 \cdot 7^2 \cdot 11 \\
&\alpha \quad 648 \\
&\beta \quad 324x^3 - 162x^2y + 162x^2 + 81xy^2 - 162xy + 81x - 27y^3 + 306y^2 + 267y + 2146 \\
&\gamma \quad 27y^3 + 369y^2 + 777y + 3563 \\
&\{i, j\} = \{12, 31\}, k_{ij} = 2^3 \cdot 3^2 \cdot 7 \\
&\alpha \quad 2697x^4 - 6728y - 9509 \\
&\beta \quad -4814x^7 - 58x^6y + 58x^6 - 522x^5y - 2146x^5 + 2755x^4y^2 + 16066x^4y \\
&\quad + 16008x^4 + 68676x^3 - 8498x^2y + 8498x^2 + 13456xy^2 - 11986xy - 105129x \\
&\quad - 1798y^3 + 36834y^2 - 10977y - 60481 \\
&\gamma \quad -12325x^6 - 4930x^5y - 7395x^5 - 58x^4y^2 - 18183x^4y - 24070x^4 \\
&\quad - 4930x^2y^2 - 51765x^2y + 109480x^2 - 4930xy^2 - 85xy + 57800x - 4930y^3 \\
&\quad - 43211y^2 + 50654y - 51476 \\
&\{i, j\} = \{12, 37\}, k_{ij} = 2^2 \cdot 3 \cdot 7 \\
&\alpha \quad -215x^4 - 970y - 2909 \\
&\beta \quad -400x^7 + 5x^6y - 5x^6 + 25x^5y - 85x^5 - 210x^4y^2 - 550x^4y - 750x^4 \\
&\quad - 5709x^3 - 51x^2y + 51x^2 - 400xy^2 + 785xy - 5891x - 385y^3 - 3170y^2 - 6407y - 9969 \\
&\gamma \quad -585x^6 - 390x^5y - 195x^5 + 5x^4y^2 + 235x^4y + 810x^4 - 585x^2y^2 \\
&\quad + 780x^2y - 9594x^2 - 585xy^2 - 5031xy - 3783x + 585y^3 - 1046y^2 + 4675y + 3572 \\
&\{i, j\} = \{12, 39\}, k_{ij} = 2^3 \cdot 3^2 \cdot 11 \\
&\alpha \quad 4x^4 - 2y - 70 \\
&\beta \quad 2x^7 - x^6y + x^6 - 8x^5y - 12x^5 + 3x^4y^2 - 12x^4y - 29x^4 - 64x^3 \\
&\quad + 32x^2y - 32x^2 + 4xy^2 + 148xy + 154x - 2y^3 - 32y^2 + 228y + 382 \\
&\gamma \quad -x^4y^2 - 14x^4y - 25x^4 + 40y^2 + 232y + 340 \\
&\{i, j\} = \{13, 17\}, k_{ij} = 7 \\
&\alpha \quad 1 \\
&\beta \quad x^3 - x^2y - 2x^2 + xy^2 + 4xy + 4x + y^3 - 5 \\
&\gamma \quad -y^3 - 3y^2 - 6y - 1 \\
&\{i, j\} = \{13, 21\}, k_{ij} = 7 \cdot 17 \\
&\alpha \quad x^4 - 19y - 71 \\
&\beta \quad x^7 - x^6y - 2x^6 - x^5y + 2x^5 - 5x^4y - 5x^4 - 62x^3 + 62x^2y + 124x^2 \\
&\quad + 19xy^2 - 5xy + 201x - 19y^3 + 48y^2 + 33y - 24 \\
&\gamma \quad -x^4y^2 - 5x^4y - 2x^4 + 81y^2 + 243y + 449 \\
&\{i, j\} = \{13, 24\}, k_{ij} = 2^2 \cdot 3^3 \\
&\alpha \quad x^4 - 4y - 5 \\
&\beta \quad x^7 - x^6y - 2x^6 - 2x^5y + 3x^5 + 2x^4y^2 + x^4y - 5x^4 + 3x^3 - 3x^2y \\
&\quad - 6x^2 + 4xy^2 - 10xy + 21x - 4y^3 + 2y^2 - 5y - 47 \\
&\gamma \quad -x^4y^2 - 6x^4y - x^4 + y^2 - 22y + 9 \\
&\{i, j\} = \{13, 26\}, k_{ij} = 2^4 \cdot 3^4 \\
&\alpha \quad x^4 - 16y - 179 \\
&\beta \quad x^7 - x^6y - 2x^6 + x^5y + 3x^5 - 2x^4y - 4x^4 - 171x^3 + 171x^2y + 342x^2 \\
&\quad + 16xy^2 - 259xy + 15x - 16y^3 + 40y^2 + 62y - 908 \\
&\gamma \quad -x^4y^2 - 3x^4y - x^4 + 187y^2 + 425y + 699 \\
&\{i, j\} = \{13, 35\}, k_{ij} = 2 \cdot 3 \cdot 17 \\
&\alpha \quad 5x^4 + 224y + 259 \\
&\beta \quad 5x^7 - 23x^6y - 46x^6 + 46x^5y + 23x^5 - 18x^4y^2 + 31x^4y + 277x^4 - 105x^3 \\
&\quad + 10833x^2y + 21666x^2 + 46xy^2 - 22172xy - 43113x + 170y^3 + 10978y^2 + 11665y - 21881 \\
&\gamma \quad -18x^5y - 36x^5 - 23x^4y^2 - 82x^4y - 141x^4 + 54x^2y^2 + 144x^2y + 54x^2 \\
&\quad + 108xy^2 + 11016xy + 21564x - 54y^3 + 11005y^2 + 43282y + 43401 \\
&\{i, j\} = \{14, 24\}, k_{ij} = 2 \cdot 3^2 \cdot 11 \\
&\alpha \quad x^4 - 6y - 7 \\
&\beta \quad -x^7 - x^6y - 2x^6 + 2x^5y - 3x^5 + 2x^4y^2 + x^4y - 7x^4 - 5x^3 - 5x^2y \\
&\quad - 10x^2 - 6xy^2 + 16xy - 45x - 6y^3 + 4y^2 - 7y - 83 \\
&\gamma \quad -x^4y^2 - 6x^4y - x^4 + y^2 - 36y + 25 \\
&\{i, j\} = \{14, 32\}, k_{ij} = 2^6 \cdot 3 \\
&\alpha \quad x^4 - 4y - 13 \\
&\beta \quad -x^7 - x^6y - 2x^6 + 3x^5y + 3x^5 + 2x^4y^2 + 2x^4y - 2x^4 + 17x^3 + 17x^2y \\
&\quad + 34x^2 - 4xy^2 - 39xy - 91x - 4y^3 - 26y^2 - 58y - 10 \\
&\gamma \quad -x^4y^2 - 7x^4y - 7x^4 + 21y^2 + 107y + 159 \\
&\{i, j\} = \{14, 35\}, k_{ij} = 2^3 \cdot 3^2 \cdot 17
\end{aligned}$$

α	$-x^4 + 24y + 60$
β	$x^7 + x^6y + 2x^6 - 2x^5y - 3x^5 + 5x^4y + 9x^4 - 36x^3 - 36x^2y - 72x^2$ $+ 24xy^2 + 24xy + 444x + 24y^3 - 48y^2 + 132y - 348$
γ	$x^4y^2 + 6x^4y + 7x^4 - 60y^2 - 168y - 588$
$\{i, j\} = \{15, 22\}, k_{ij} = 11^2 \cdot 13$	
α	$583x^4 - 9381y + 11919$
β	$901x^7 + 159x^6y - 318x^6 - 477x^5y - 1272x^5 + 424x^4y^2 - 4187x^4y$ $+ 901x^4 + 12230x^3 + 22548x^2y - 45096x^2 + 8427xy^2 - 26463xy + 22265x - 3445y^3$ $- 17254y^2 - 5477y + 6746$
γ	$-1484x^6 + 742x^5y - 2968x^5 + 159x^4y^2 - 901x^4y - 1060x^4 - 5936x^2y^2$ $+ 14098x^2y - 25368x^2 - 5936xy^2 + 3780xy - 4732x - 5936y^3 + 22283y^2 - 85317y$ $+ 63195$
$\{i, j\} = \{15, 26\}, k_{ij} = 2^3 \cdot 5 \cdot 13$	
α	$x^4 - 18y + 49$
β	$-x^7 - x^6y + 2x^6 + x^5y + 3x^5 + 4x^4y - 14x^4 - 27x^3 - 27x^2y + 54x^2$ $- 18xy^2 - 23xy - 155x - 18y^3 + 4y^2 + 40y + 308$
γ	$-x^4y^2 + 3x^4y - 7x^4 - 9y^2 + 131y + 47$
$\{i, j\} = \{15, 32\}, k_{ij} = 2^6 \cdot 3 \cdot 7$	
α	$x^4 - 48y + 187$
β	$-x^7 - x^6y + 2x^6 + 3x^5y + 7x^5 + 2x^4y^2 + 2x^4y - 26x^4 - 123x^3$ $- 123x^2y + 246x^2 - 48xy^2 + 145xy - 483x - 48y^3 + 166y^2 - 378y + 770$
γ	$-x^4y^2 + x^4y - 11x^4 - 75y^2 + 347y - 9$
$\{i, j\} = \{15, 39\}, k_{ij} = 2 \cdot 7^2$	
α	$-x^4 - 2y - 21$
β	$x^7 + x^6y - 2x^6 + 2x^5y - 13x^5 - 5x^4y - 7x^4 + 17x^3 + 17x^2y - 34x^2$ $- 2xy^2 + 28xy - 239x - 2y^3 - 6y^2 - 101y - 115$
γ	$x^4y^2 - 6x^4y + 17x^4 + 19y^2 - 96y + 307$
$\{i, j\} = \{16, 21\}, k_{ij} = 2$	
α	$91x^4 + 962y - 200$
β	$-260x^7 + 13x^6y - 13x^6 - 65x^5y + 169x^5 + 104x^4y^2 + 208x^4y - 39x^4$ $+ 1265x^3 - 1364x^2y + 1364x^2 + 1352xy^2 + 1100xy - 167x + 819y^3 - 1733y^2 + 1201y$ $- 198$
γ	$429x^6 - 286x^5y - 143x^5 + 13x^4y^2 - 39x^4y - 182x^4 + 143x^2y^2$ $- 1573x^2y - 2200x^2 - 143xy^2 + 5566xy - 1793x - 143y^3 - 2352y^2 - 2838y + 763$
$\{i, j\} = \{16, 26\}, k_{ij} = 2^5 \cdot 3^2 \cdot 5$	
α	$-28x^4 + 196y + 44$
β	$14x^7 + 7x^6y - 7x^6 - 49x^5y - 21x^5 - 21x^4y^2 + 56x^4y + 49x^4 - 36x^3$ $- 18x^2y + 18x^2 + 392xy^2 + 70xy + 2994x + 196y^3 - 562y^2 + 1312y - 4522$
γ	$7x^4y^2 + 84x^4y + 49x^4 - 802y^2 - 104y - 6006$
$\{i, j\} = \{16, 27\}, k_{ij} = 2^4$	
α	$-4x^4 + 4y$
β	$2x^7 + x^6y - x^6 - 7x^5y + x^5 - 3x^4y^2 + 10x^4y + x^4 - 2x^3 - x^2y + x^2$ $+ 8xy^2 - xy - 5x + 4y^3 - 13y^2 - 2y + 7$
γ	$x^4y^2 + 12x^4y - x^4 - 17y^2 + 4y + 9$
$\{i, j\} = \{16, 28\}, k_{ij} = 2^3 \cdot 3^5$	
α	$28x^4 - 98y + 194$
β	$-14x^7 - 7x^6y + 7x^6 + 42x^5y - 84x^5 + 21x^4y^2 - 63x^4y + 28x^4 - 258x^3$ $- 129x^2y + 129x^2 - 196xy^2 + 522xy - 554x - 98y^3 + 359y^2 - 489y + 180$
γ	$-7x^4y^2 - 70x^4y + 161x^4 + 263y^2 - 786y + 979$
$\{i, j\} = \{16, 37\}, k_{ij} = 2^4 \cdot 3^3$	
α	$4x^4 + 8y + 4$
β	$-2x^7 - x^6y + x^6 + 7x^5y - 3x^5 + 3x^4y^2 - 11x^4y + 2x^4 - 6x^3 - 3x^2y$ $+ 3x^2 + 16xy^2 + 5xy - 81x + 8y^3 - 23y^2 - 49y + 118$
γ	$-x^4y^2 - 12x^4y + 5x^4 - 35y^2 - 4y + 159$
$\{i, j\} = \{17, 23\}, k_{ij} = 2^2 \cdot 7^2 \cdot 13$	
α	$22578x^4 - 226098y - 20030$
β	$-43566x^7 - 159x^6y + 318x^6 - 1590x^5y - 11766x^5 + 22737x^4y^2 - 17649x^4y$ $- 44361x^4 - 107056x^3 - 88543x^2y + 177086x^2 + 101124xy^2 - 75166xy + 121930x$ $- 50562y^3 + 280778y^2 - 391251y - 172227$
γ	$-109710x^6 - 43884x^5y + 87768x^5 - 159x^4y^2 - 112254x^4y - 133878x^4$

$$\begin{aligned}
& -175536x^2y^2 - 164565x^2y - 216660x^2 - 284142xy + 568284x - 175536y^3 \\
& -226396y^2 - 177385y - 326972 \\
\{i, j\} = \{17, 27\}, k_{ij} = 2^2 \cdot 7 \cdot 11 \\
\alpha & 8x^4 - 8y - 60 \\
\beta & 4x^7 - 2x^6y + 4x^6 - 10x^5y - 4x^5 + 6x^4y^2 - 2x^4y + 8x^4 - 34x^3 \\
& + 17x^2y - 34x^2 + 16xy^2 + 37xy + 154x - 8y^3 - 27y^2 - 47y - 64 \\
\gamma & -2x^4y^2 - 12x^4y - 16x^4 + 49y^2 + 6y + 376 \\
\{i, j\} = \{17, 29\}, k_{ij} = 2^2 \cdot 3 \cdot 11 \cdot 13 \\
\alpha & 16x^4 - 96y + 60 \\
\beta & 8x^7 - 4x^6y + 8x^6 - 28x^5y - 72x^5 + 12x^4y^2 - 12x^4y - 144x^4 - 42x^3 \\
& + 21x^2y - 42x^2 + 192xy^2 + 243xy + 858x - 96y^3 + 273y^2 - 33y + 1788 \\
\gamma & -4x^4y^2 - 40x^4y - 160x^4 + 405y^2 + 402y + 1800 \\
\{i, j\} = \{17, 35\}, k_{ij} = 2^2 \cdot 3 \\
\alpha & 20x^4 + 558y - 816 \\
\beta & -74x^7 - 19x^6y + 38x^6 - 76x^5y - 114x^5 + x^4y^2 + 113x^4y + 648x^4 \\
& + 1978x^3 - 513x^2y + 1026x^2 + 228xy^2 - 1596xy - 7032x + 334y^3 + 831y^2 + 3759y \\
& - 3364 \\
\gamma & -168x^6 - 112x^5y - 112x^5 - 19x^4y^2 - 132x^4y - 696x^4 + 224x^2y^2 \\
& + 308x^2y + 2800x^2 + 448xy^2 + 1568xy + 3696x - 224y^3 + 755y^2 - 496y - 5924 \\
\{i, j\} = \{17, 37\}, k_{ij} = 2^2 \\
\alpha & -4 \\
\beta & -2x^3 + x^2y - 2x^2 + 5xy - 3y^2 + 2y - 4 \\
\gamma & y^2 + 6y + 4 \\
\{i, j\} = \{18, 23\}, k_{ij} = 2^2 \cdot 13 \\
\alpha & 68x^4 - 578y - 16 \\
\beta & -34x^7 - 17x^6y - 34x^6 + 170x^5y + 170x^5 + 85x^4y^2 + 255x^4y + 136x^4 \\
& - 26x^3 - 13x^2y - 26x^2 - 1156xy^2 - 6xy - 7384x - 578y^3 - 1159y^2 - 3409y - 7376 \\
\gamma & -17x^4y^2 - 408x^4y - 408x^4 + 2299y^2 - 40y + 14716 \\
\{i, j\} = \{19, 35\}, k_{ij} = 3^3 \cdot 7 \\
\alpha & 64y - 48 \\
\beta & -32x^7 + 64x^6 + 32x^5 - 80x^4y - 168x^3 + 64x^2y^2 - 48x^2y + 384x^2 - 66x \\
& - 64y^3 + 48y^2 - 500y + 609 \\
\gamma & 32x^7 + 16x^4y + 168x^3 + 234x + 116y - 141 \\
\{i, j\} = \{20, 31\}, k_{ij} = 2 \cdot 3 \\
\alpha & -y + 2 \\
\beta & x^7 - x^6 - x^5 - 2x^4y + 4x^3 - x^2y^2 + 2x^2y - 6x^2 - 14x - y^3 + 2y^2 - 9y \\
& + 18 \\
\gamma & -x^7 + x^4y - 4x^3 + 10x + 3y - 8 \\
\{i, j\} = \{20, 34\}, k_{ij} = 2^3 \\
\alpha & -y + 2 \\
\beta & -x^7 - x^6 + x^5 - 2x^4y - 4x^3 - x^2y^2 + 2x^2y - 6x^2 + 12x - y^3 + 2y^2 - 11y \\
& + 18 \\
\gamma & x^7 + x^4y + 4x^3 - 8x + 5y - 10 \\
\{i, j\} = \{21, 35\}, k_{ij} = 11 \cdot 17 \\
\alpha & -2187y + 8505 \\
\beta & 729x^7 - 2187x^6 + 1458x^5 + 972x^4y - 972x^4 - 1566x^3 - 2187x^2y^2 \\
& + 4860x^2y - 4293x^2 + 2187xy^2 - 4860xy + 20220x + 729y^3 - 3078y^2 + 4878y - 40250 \\
\gamma & -729x^7 + 1215x^4y - 486x^4 + 1566x^3 - 14361x + 1458y^3 - 3969y^2 + 4275y - 8692 \\
\{i, j\} = \{22, 25\}, k_{ij} = 2 \\
\alpha & -2y - 2 \\
\beta & -x^7 - 2x^6 - 2x^5 - x^4 - 7x^3 - 2x^2y^2 - 12x^2 - 2xy^2 - 11x - y^2 - y - 6 \\
\gamma & x^7 - 2x^4y - x^4 + 7x^3 - x - 2y^3 - y^2 - 11y - 4 \\
\{i, j\} = \{22, 29\}, k_{ij} = 5^2 \\
\alpha & -16x^4 + 16y - 12 \\
\beta & 8x^7 + 32x^6 + 16x^5y + 24x^5 - 8x^4y^2 + 12x^4y + 4x^4 - 52x^3 - 16x^2y^2 \\
& + 100x^2y - 88x^2 - 16xy^2 - 4xy - 38x + 44y^2 - 18y - 27 \\
\gamma & -8x^7 + 8x^4y^2 + 36x^4y + 20x^4 + 52x^3 + 2x - 16y^3 + 40y^2 - 74y - 13 \\
\{i, j\} = \{22, 32\}, k_{ij} = 1 \\
\alpha & 0
\end{aligned}$$

β	-1
γ	1
$\{i, j\}$	$\{23, 29\}, k_{ij} = 2^2 \cdot 5 \cdot 13$
α	$84x^4 - 21y - 155$
β	$-21x^7 - 105x^6 - 42x^5y + 42x^4y^2 - 63x^4y - 21x^4 - 103x^3 - 21x^2y^2$ $-525x^2y + 31x^2 - 206xy - 136x - 21y^3 - 340y^2 - 37y - 260$
γ	$21x^7 - 42x^4y^2 - 42x^4y - 84x^4 + 103x^3 + 136x - 206y^2 - 457y + 155$
$\{i, j\}$	$\{25, 33\}, k_{ij} = 3 \cdot 5 \cdot 11$
α	$y + 3$
β	$-x^7 + x^6 + x^5 + x^4y - 19x^3 + x^2y^2 - 2x^2y + 14x^2 - xy^2 + 2xy - 39x$ $-y^3 + 2y^2 + 23y + 123$
γ	$x^7 - x^4 + 19x^3 + 63x + 2y^3 - 3y^2 - 11y - 84$
$\{i, j\}$	$\{26, 27\}, k_{ij} = 2$
α	0
β	-1
γ	1
$\{i, j\}$	$\{26, 28\}, k_{ij} = 2^3 \cdot 3 \cdot 5 \cdot 11$
α	-1
β	$x^5 + x^4 + 14x^3 + 4x^2y + 7x^2 + 144x + 41y - 351$
γ	$-x^5 - 10x^3 - 5x^2y + 9x^2 - 103x + y^2 - 43y + 484$
$\{i, j\}$	$\{27, 32\}, k_{ij} = 2$
α	-2
β	$2x^2 + y^2 - y + 7$
γ	$2x - y^2 - y - 7$
$\{i, j\}$	$\{28, 32\}, k_{ij} = 2^6 \cdot 3^3 \cdot 5$
α	$-484x^4 - 1089y - 752$
β	$121x^7 + 363x^6 - 242x^5y - 121x^5 + 242x^4y^2 + 242x^4y - 54x^3$ $-1089x^2y^2 + 3388x^2y + 1048x^2 + 108xy - 3758x + 1089y^3 - 1318y^2 + 619y + 6172$
γ	$-121x^7 - 242x^4y^2 - 605x^4y - 726x^4 + 54x^3 + 3812x + 108y^2 - 8443y - 4456$
$\{i, j\}$	$\{29, 38\}, k_{ij} = 2^2 \cdot 3^2$
α	$-y + 7$
β	$-x^7 - x^6 + x^5 + 3x^4y - 5x^4 + 17x^3 - x^2y^2 + 4x^2y + 8x^2 + xy^2 - 4xy$ $-67x + 3y^3 - 17y^2 - 5y - 71$
γ	$x^7 - 2x^4y + 3x^4 - 17x^3 + 59x - 2y^3 + 11y^2 + 5y + 28$
$\{i, j\}$	$\{30, 39\}, k_{ij} = 1$
α	0
β	-1
γ	1
$\{i, j\}$	$\{31, 35\}, k_{ij} = 2^4 \cdot 3^4 \cdot 7$
α	$-460x^4 + 8625y - 887$
β	$-115x^7 + 575x^6 - 230x^5y + 575x^5 - 230x^4y^2 + 1035x^4y + 3335x^4$ $-4103x^3 + 1725x^2y^2 + 5680x^2 + 1725xy^2 - 8206xy + 25225x + 5175y^3 + 419y^2$ $+22887y - 9803$
γ	$115x^7 + 230x^4y^2 - 690x^4y - 2185x^4 + 4103x^3 - 19545x - 3450y^3 + 3031y^2$ $-25413y + 1618$
$\{i, j\}$	$\{31, 39\}, k_{ij} = 2^4 \cdot 3^4 \cdot 5 \cdot 7$
α	$11628x^4 - 13889y - 19559$
β	$-8721x^7 - 37791x^6 - 17442x^5y - 55233x^5 + 5814x^4y^2 - 12597x^4y$ $-39729x^4 + 7321x^3 + 41667x^2y^2 - 12274x^2y + 66716x^2 + 41667xy^2 + 2368xy$ $+105001x + 13889y^3 + 32695y^2 - 10427y + 96601$
γ	$8721x^7 - 5814x^4y^2 - 42636x^4y - 53295x^4 - 7321x^3 - 23643x + 27778y^3$ $+38365y^2 + 67237y + 122402$
$\{i, j\}$	$\{33, 34\}, k_{ij} = 2^3 \cdot 5$
α	$-y - 6$
β	$-x^7 - x^6 - x^5 + 2x^3 - x^2y^2 - 2x^2y + 6x^2 - 46x - y^3 - 4y^2 - y + 2$
γ	$x^7 - x^4y - 2x^4 - 2x^3 + 48x + 3y - 38$
$\{i, j\}$	$\{34, 36\}, k_{ij} = 2^4 \cdot 5^4$
α	$4644x^4 - 5547y - 6797$
β	$3483x^7 - 15093x^6 + 6966x^5y + 22059x^5 + 2322x^4y^2 - 5031x^4y - 18963x^4$

	$-3711x^3 + 16641x^2y^2 - 11094x^2y + 21800x^2 - 16641xy^2 + 3672xy - 45003x$
	$+5547y^3 + 10469y^2 - 12499y + 47581$
γ	$-3483x^7 - 2322x^4y^2 - 17028x^4y - 18189x^4 + 3711x^3 + 15781x + 11094y^3$
	$+11719y^2 + 19533y + 50784$
	$\{i, j\} = \{35, 38\}, k_{ij} = 2^5 \cdot 3^2$
α	$-4y + 52$
β	$-x^7 - 4x^6 - 8x^5 - 8x^4 + 59x^3 - 4x^2y^2 + 28x^2y + 164x^2 - 8xy^2 + 56xy$
	$+38x - 8y^2 + 55y - 239$
γ	$x^7 + 4x^4y - 12x^4 - 59x^3 + 290x + 4y^3 - 40y^2 - 79y - 101$

Table 3: The polynomials $\varphi_i(x) = k_i^3 F(x, -r_i(x)/k_i)$ of Step 2 and the signs of H_i for $x \geq 10$ and for $x \leq -11$.

i	k_i	$\varphi_i(x)$	$x \geq 10$	$x \leq -11$
0	-1	$-x^4 - x + 1$	+	+
1	-1	$-x^4 + x^3 + 5x + 1$	+	+
2	-1	$-x^4 - x + 8$	+	+
3	-1	$-x^4 - x - 6$	+	+
4	-1	$-x^4 - x^3 - 7x + 1$	+	+
5	-1	$-x^4 + 8x^3 + 11x + 1$	+	+
6	-1	$-x^4 + x^3 + 3x^2 + 8x + 8$	+	+
7	-1	$-x^4 + x^3 - 3x^2 + 8x - 6$	+	+
8	-1	$-x^4 - x + 21$	+	+
9	-1	$-x^4 - x^3 + 3x^2 - 10x + 8$	+	+
10	-1	$-x^4 - 8x^3 - 13x + 1$	+	+
11	-1	$-x^4 + 8x^3 + 12x^2 + 17x + 8$	+	+
12	-1	$-x^4 + 8x^3 - 12x^2 + 17x - 6$	+	+
13	-1	$-x^4 + x^3 + 6x^2 + 17x + 21$	+	+
14	-1	$-x^4 - x^3 + 6x^2 - 19x + 21$	+	+
15	-1	$-x^4 - x^3 - 6x^2 - 19x - 19$	+	+
16	-1	$-x^4 - 8x^3 - 12x^2 - 19x - 6$	+	+
17	-1	$-x^4 + 8x^3 - 24x^2 + 35x - 19$	+	+
18	-1	$-x^4 - 8x^3 + 24x^2 - 37x + 21$	+	+
19	1	$-x^6 + x^4 - 6x^2 + x - 1$	+	+
20	-1	$-x^6 - x^4 - 6x^2 - x + 1$	+	+
21	1	$-x^6 - 3x^5 - 2x^4 - x^3 - 6x^2 - 5x - 1$	+	+
22	-1	$-x^6 + 3x^5 - 4x^4 + x^3 - 6x^2 + 5x + 1$	+	+
23	-1	$-x^6 + 2x^4 - 9x^2 - x + 8$	+	+
24	-1	$-x^6 - 4x^4 - 9x^2 - x - 6$	+	+
25	-1	$-x^6 - 3x^5 - 4x^4 - x^3 - 6x^2 - 7x + 1$	+	+
26	1	$-x^6 + 3x^5 + x^4 - 5x^3 - 6x^2 + 10x + 6$	+	+
27	1	$-x^6 + 3x^5 - 5x^4 + 7x^3 - 12x^2 + 10x - 8$	+	+
28	1	$-x^6 - 5x^4 - 18x^2 + x - 21$	+	+
29	1	$-x^6 - 3x^5 + x^4 + 5x^3 - 6x^2 - 8x + 6$	+	+
30	1	$-x^6 - 6x^5 - 11x^4 - 8x^3 - 6x^2 - 11x - 1$	+	+
31	-1	$-x^6 + 3x^5 - x^4 - 5x^3 - 6x^2 + 8x + 8$	+	+
32	-1	$-x^6 + 3x^5 - 7x^4 + 7x^3 - 12x^2 + 8x - 6$	+	+
33	-1	$-x^6 + 5x^4 - 18x^2 - x + 21$	+	+
34	-1	$-x^6 - 3x^5 - x^4 + 5x^3 - 6x^2 - 10x + 8$	+	+
35	1	$-x^6 + 6x^5 - 8x^4 - 4x^3 + 3x^2 + 19x + 6$	+	+
36	1	$-x^6 + 6x^5 - 14x^4 + 20x^3 - 21x^2 + 19x - 8$	+	+
37	1	$-x^6 + 3x^5 - 8x^4 + 13x^3 - 24x^2 + 19x - 21$	+	+
38	1	$-x^6 - 6x^5 - 8x^4 + 4x^3 + 3x^2 - 17x + 6$	+	+
39	1	$-x^6 - 6x^5 - 14x^4 - 20x^3 - 21x^2 - 17x - 8$	+	+

Table 4: The terminal residue classes $x \equiv a, y \equiv b \pmod{p^k}$ of Step 4, grouped by p, k and the value of B_p .

p	k	B_p	classes (a, b)
2	6	1	(0, 7), (1, 57), (2, 41), (3, 3), (4, 19), (11, 27), (12, 11), (13, 13), (14, 13), (15, 55), (16, 55), (17, 41), (18, 25), (19, 51), (20, 3), (27, 11), (28, 59), (30, 61), (31, 39), (32, 39), (33, 25), (34, 9), (35, 35), (36, 51), (43, 59), (44, 43), (46, 45), (47, 23), (48, 23), (49, 9), (50, 57), (51, 19), (52, 35), (59, 43), (60, 27), (62, 29), (63, 7)
2	7	1	(6, 85), (7, 31), (8, 127), (9, 81), (10, 97), (22, 69), (23, 79), (24, 111), (25, 1), (26, 81), (29, 61), (38, 53), (39, 127), (40, 95), (41, 49), (42, 65), (45, 109), (54, 37), (55, 47), (56, 79), (57, 97), (58, 49), (61, 29), (70, 21), (71, 95), (72, 63), (73, 17), (74, 33), (86, 5), (87, 15), (88, 47), (89, 65), (90, 17), (93, 125), (102, 117), (103, 63), (104, 31), (105, 113), (106, 1), (109, 45), (118, 101), (119, 111), (120, 15), (121, 33), (122, 113), (125, 93)
2	8	1	(5, 245), (21, 165), (37, 85), (69, 181), (101, 21), (133, 117), (165, 213), (181, 133), (197, 53), (229, 149)
2	9	1	(53, 261), (85, 101), (117, 453), (149, 293), (213, 485), (245, 325), (309, 5), (341, 357), (373, 197), (405, 37), (469, 229), (501, 69)
3	3	1	(13, 2), (13, 5), (13, 8), (13, 11), (13, 14), (13, 17), (13, 23), (13, 26), (21, 1), (21, 4), (21, 7), (21, 10), (21, 13), (21, 19), (21, 22), (21, 25)
3	4	1	(13, 20), (13, 47), (13, 74), (21, 16), (21, 70)
3	5	1	(102, 43), (102, 124), (102, 205)
5	2	-1	(2, 9), (7, 24), (12, 14), (17, 4), (23, 21)
5	3	-1	(3, 41), (8, 11), (13, 81), (18, 1), (22, 19), (28, 16), (33, 111), (38, 56), (43, 101), (47, 94), (53, 116), (58, 86), (68, 76), (72, 44), (78, 91), (83, 61), (88, 6), (93, 51), (97, 119), (103, 66), (108, 36), (113, 106), (118, 26), (122, 69)
5	4	-1	(63, 531), (188, 406), (313, 281), (438, 156), (563, 31)
7	1	1	(0, 5), (1, 2), (6, 5)
7	2	1	(2, 32), (3, 19), (4, 14), (4, 20), (5, 5), (9, 25), (10, 12), (11, 0), (11, 27), (17, 5), (18, 34), (18, 35), (19, 40), (23, 11), (24, 47), (25, 21), (25, 36), (25, 41), (26, 33), (30, 4), (31, 40), (32, 7), (32, 43), (32, 48), (33, 26), (37, 46), (38, 33), (39, 6), (39, 42), (40, 19), (44, 39), (45, 26), (46, 13), (46, 28), (47, 12)
7	3	1	(4, 211), (11, 169), (12, 47), (16, 263), (18, 78), (39, 197), (46, 253), (53, 260), (60, 218), (61, 341), (65, 214), (67, 127), (88, 246), (95, 302), (102, 309), (109, 267), (110, 292), (114, 165), (116, 176), (137, 295), (144, 8), (151, 15), (158, 316), (159, 243), (163, 116), (165, 225), (186, 1), (193, 57), (200, 64), (207, 22), (208, 194), (212, 67), (214, 274), (235, 50), (242, 106), (249, 113), (256, 71), (257, 145), (261, 18), (263, 323), (284, 99), (291, 155), (298, 162), (305, 120), (306, 96), (310, 312), (312, 29), (333, 148), (340, 204)
11	1	1	(0, 3), (0, 5), (1, 6), (1, 8), (5, 2), (8, 4), (10, 3), (10, 5)
11	2	1	(7, 20), (8, 18), (8, 22), (9, 46), (18, 75), (19, 51), (19, 77), (20, 112), (29, 9), (30, 11), (30, 84), (31, 57), (40, 64), (41, 66), (41, 117), (42, 2), (51, 119), (52, 0), (52, 29), (53, 68), (62, 53), (63, 55), (63, 62), (64, 13), (73, 108), (74, 95), (74, 110), (75, 79), (84, 42), (85, 7), (85, 44), (86, 24), (96, 40), (96, 99), (97, 90), (106, 31), (107, 33), (107, 73), (108, 35), (117, 86), (118, 88), (118, 106), (119, 101)
11	3	1	(95, 1186), (216, 460), (337, 1065), (458, 339), (579, 944), (700, 218), (821, 823), (942, 97), (1063, 702), (1184, 1307), (1305, 581)
13	1	1	(0, 8), (1, 5), (3, 7), (6, 5), (8, 6), (11, 0), (12, 8)

p	k	B_p	classes (a, b)
13	2	1	(4, 73), (5, 7), (10, 34), (17, 86), (18, 150), (23, 47), (30, 99), (31, 124), (36, 60), (43, 112), (44, 98), (49, 73), (56, 125), (57, 72), (62, 86), (70, 46), (75, 99), (82, 151), (83, 20), (88, 112), (95, 164), (96, 163), (101, 125), (108, 8), (109, 137), (114, 138), (121, 21), (122, 111), (127, 151), (134, 34), (135, 85), (140, 164), (147, 47), (148, 59), (153, 8), (160, 60), (161, 33), (166, 21)
13	3	1	(69, 1659), (238, 1828), (407, 1997), (576, 2166), (745, 138), (914, 307), (1083, 476), (1252, 645), (1421, 814), (1590, 983), (1759, 1152), (1928, 1321), (2097, 1490)
17	1	1	(0, 9), (1, 8), (2, 0), (3, 5), (5, 0), (7, 1), (8, 3), (8, 7), (9, 16), (11, 2), (11, 11), (12, 16), (16, 9)
17	2	1	(11, 4), (28, 174), (45, 55), (62, 225), (79, 106), (96, 276), (113, 157), (130, 38), (147, 208), (164, 89), (181, 259), (198, 140), (215, 21), (232, 191), (249, 72), (266, 242), (283, 123)